Column Space, Null Space and Rank-Nullity

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Column space and null space

Definition 0.1 (Column space, range, image)**.** The **column space** [1] of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{C}(\mathbf{A}) \stackrel{\text{def}}{=} \{\mathbf{A}\underset{\sim}{x} : \underset{\sim}{x} \in \mathbb{R}^n\},$$

the set of all vectors $\mathbf{A}\underset{\sim}{x}$ in $\mathbb{R}^m$. It is also called the **range** or **image** of $\mathbf{A}$.

Definition 0.2 (Row space)**.** The **row space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{C}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^n$.

Example 0.1 (The column space of a rank-one matrix)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}.$$

For any $\underset{\sim}{x} \in \mathbb{R}^3$,

$$\begin{aligned} \mathbf{A}\underset{\sim}{x} &= (x_1 - 2x_2 - 2x_3,\ 3\,(x_1 - 2x_2 - 2x_3)) \\ &= (x_1 - 2x_2 - 2x_3)\,(1, 3) \end{aligned}$$

(Definition 16 in Matrices). The number $x_1 - 2x_2 - 2x_3$ can be any real number (take $x_2 = 0$ and $x_3 = 0$), so $\mathcal{C}(\mathbf{A}) = \{c\,(1, 3) : c \in \mathbb{R}\}$, a line in $\mathbb{R}^2$. The vector $(1, 0)$ is not in $\mathcal{C}(\mathbf{A})$: $c\,(1, 3)$ has second entry $3c$, which is 0 only if $c = 0$, and then the first entry is 0, not 1.

Theorem 0.1 (The column space is the span of the columns)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix with columns* $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$*, then*

$$\mathcal{C}(\mathbf{A}) = \operatorname{span}\{\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n\},$$

so $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$, and the row space $\mathcal{C}(\mathbf{A}^\top)$ is a subspace of $\mathbb{R}^n$.

Proof

Proof. By Theorem 3 in Subspaces and Rank, $\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n$, and as $\underset{\sim}{x}$ ranges over $\mathbb{R}^n$, the coefficients $x_1, \ldots, x_n$ range over all lists of n numbers. So the set of all $\mathbf{A}\underset{\sim}{x}$ (Definition 0.1) is the set of all linear combinations of the columns, which is their span (Definition 5 in Subspaces and Rank). A span is a subspace (Theorem 2 in Subspaces and Rank). Applying the same argument to $\mathbf{A}^\top$, whose columns are the rows of $\mathbf{A}$, shows the row space is a subspace of $\mathbb{R}^n$. □

Example 0.2 (Reading off the column space and row space from the columns and rows)**.** For $\mathbf{A}$ in Example 0.1, the columns are $(1, 3)$, $(-2, -6) = -2\,(1, 3)$ and $(-2, -6) = -2\,(1, 3)$, so by Theorem 0.1 $\mathcal{C}(\mathbf{A}) = \operatorname{span}\{(1, 3)\}$, the same line found in Example 0.1. The rows are $(1, -2, -2)$ and $(3, -6, -6) = 3\,(1, -2, -2)$, so the row space is $\operatorname{span}\{(1, -2, -2)\}$, a line in $\mathbb{R}^3$.

Definition 0.3 (Null space, kernel)**.** The **null space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^n : \mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m\},$$

the set of solutions of the linear system $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$. It is also called the **kernel** of $\mathbf{A}$.

Definition 0.4 (Left null space)**.** The **left null space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{N}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^m$.

Example 0.3 (The null space and left null space of a rank-one matrix)**.** For $\mathbf{A}$ in Example 0.1, $\mathbf{A}\underset{\sim}{x} = (x_1 - 2x_2 - 2x_3,\ 3\,(x_1 - 2x_2 - 2x_3))$, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_2$ exactly when $x_1 = 2x_2 + 2x_3$. Writing

$$\begin{aligned}\underset{\sim}{x} &= (2x_2 + 2x_3, x_2, x_3)\\ &= x_2\,(2, 1, 0) + x_3\,(2, 0, 1)\end{aligned}$$

shows $\mathcal{N}(\mathbf{A}) = \operatorname{span}\{(2, 1, 0), (2, 0, 1)\}$, a plane in $\mathbb{R}^3$.
For the left null space, $\mathbf{A}^\top y = (y_1 + 3y_2,\ -2\,(y_1 + 3y_2),\ -2\,(y_1 + 3y_2))$, which is $\underset{\sim}{0}_3$ exactly when $y_1 = -3y_2$, so $\mathcal{N}(\mathbf{A}^\top) = \operatorname{span}\{(-3, 1)\}$, a line in $\mathbb{R}^2$.

Exercise 0.1 (Testing whether a vector is in a column space)**.** A vector $\underset{\sim}{b}$ is in the column space of a matrix $\mathbf{A}$ exactly when some $\underset{\sim}{x}$ gives $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}.$$

Decide whether $\underset{\sim}{b}$ is in $\mathcal{C}(\mathbf{A})$.

Hint

Try $\underset{\sim}{x} = (x_1, 0)$ and solve for x_1.

Solution

Solution 0.1. Take $\underset{\sim}{x} = (2, 0)$. Then

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 0 \\ 3 \cdot 2 + 6 \cdot 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \underset{\sim}{b},$$

so $\underset{\sim}{b}$ is in $\mathcal{C}(\mathbf{A})$.

Exercise 0.2 (Testing whether a vector is in a row space)**.** The row space of a matrix $\mathbf{A}$ is the column space of its transpose. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

and let $\underset{\sim}{v}$ be the column vector $(1, 2)$.
Decide whether $\underset{\sim}{v}$ is in the row space $\mathcal{C}(\mathbf{A}^\top)$.

Hint

Write out $\mathbf{A}^\top$ and look for $\underset{\sim}{u}$ with $\mathbf{A}^\top\underset{\sim}{u} = \underset{\sim}{v}$.

Solution

Solution 0.2. Here

$$\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

Take $\underset{\sim}{u} = (1, 0)$. Then

$$\mathbf{A}^\top\underset{\sim}{u} = \begin{bmatrix} 1 \cdot 1 + 3 \cdot 0 \\ 2 \cdot 1 + 6 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \underset{\sim}{v},$$

so $\underset{\sim}{v}$ is in the row space.

Exercise 0.3 (Showing that a column space is all of $\mathbb{R}^2$)**.** Theorem 0.1 describes a column space as the span of the columns of $\mathbf{A}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Show that $\mathcal{C}(\mathbf{A}) = \mathbb{R}^2$.

Hint

Two of the columns of $\mathbf{A}$ are the indicator vectors $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$.

Solution

Solution 0.3. By Theorem 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1,0),(0,1),(1,1)\}$. Every $(b_1, b_2) \in \mathbb{R}^2$ is $b_1\,(1,0) + b_2\,(0,1) + 0\,(1,1)$, so it is in that span. The column space also lies inside $\mathbb{R}^2$, so $\mathcal{C}(\mathbf{A}) = \mathbb{R}^2$.

Exercise 0.4 (Finding a null space)**.** The null space of a matrix $\mathbf{A}$ holds every $\underset{\sim}{x}$ with $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Find $\mathcal{N}(\mathbf{A})$ as a span.

Hint

Write out the two entries of $\mathbf{A}\underset{\sim}{x}$ and make each 0.

Solution

Solution 0.4. For $\underset{\sim}{x} = (x_1, x_2, x_3)$,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} x_1 + 2x_2 \\ x_3 \end{bmatrix}.$$

This vector is $\underset{\sim}{0}_2$ exactly when $x_3 = 0$ and $x_1 = -2x_2$. So $\underset{\sim}{x} = (-2x_2, x_2, 0) = x_2\,(-2,1,0)$, and

$$\mathcal{N}(\mathbf{A}) = \text{span}\{(-2,1,0)\}.$$

Exercise 0.5 (Testing whether a vector is in a left null space)**.** The left null space of a matrix $\mathbf{A}$ is the null space of its transpose. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

and let y be the column vector $(2, -1)$.
Decide whether $\underset{\sim}{y}$ is in $\mathcal{N}(\mathbf{A}^\top)$.

Hint

Compute $\mathbf{A}^\top \underset{\sim}{y}$.

Solution

Solution 0.5. Here

$$\mathbf{A}^\top \underset{\sim}{y} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2-2 \\ 4-4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

so $\underset{\sim}{y}$ is in $\mathcal{N}(\mathbf{A}^\top)$. Here $\mathbf{A}$ equals its own transpose, so $\underset{\sim}{y}$ is in $\mathcal{N}(\mathbf{A})$ as well.

Theorem 0.2 (A null space is a subspace)**.** *For any $m \times n$ matrix $\mathbf{A}$, $\mathcal{N}(\mathbf{A})$ (Definition 0.3) is a subspace of $\mathbb{R}^n$ (Definition 4 in Subspaces and Rank).*

Proof

Proof. $\mathbf{A}\underset{\sim}{0}_n = \underset{\sim}{0}_m$, so $\underset{\sim}{0}_n \in \mathcal{N}(\mathbf{A})$, and the null space is not empty. If $\underset{\sim}{u}, \underset{\sim}{w} \in \mathcal{N}(\mathbf{A})$ and $c \in \mathbb{R}$, then

$$\begin{aligned} \mathbf{A}\left(\underset{\sim}{u} + \underset{\sim}{w}\right) &= \mathbf{A}\underset{\sim}{u} + \mathbf{A}\underset{\sim}{w} && \text{(Theorem 8 in Matrices)} \\ &= \underset{\sim}{0}_m + \underset{\sim}{0}_m && (\underset{\sim}{u} \text{ and } \underset{\sim}{w} \text{ are in the null space}) \\ &= \underset{\sim}{0}_m && (\text{adding } \underset{\sim}{0}_m \text{ changes nothing}) \end{aligned}$$

and, for each entry $i = 1, \ldots, m$,

$$\begin{aligned} \left(\mathbf{A}\left(c\underset{\sim}{u}\right)\right)_i &= a_{i1}(cu_1) + \cdots + a_{in}(cu_n) && \text{(Definition 16 in Matrices)} \\ &= c\left(a_{i1}u_1 + \cdots + a_{in}u_n\right) && (\text{factor } c \text{ out of each term}) \\ &= c\left(\mathbf{A}\underset{\sim}{u}\right)_i && \text{(Definition 16 in Matrices)} \\ &= c \cdot 0 && (\underset{\sim}{u} \text{ is in the null space}) \\ &= 0, && (\text{arithmetic}) \end{aligned}$$

so $\mathbf{A}\left(c\underset{\sim}{u}\right) = \underset{\sim}{0}_m$. Therefore $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\underset{\sim}{u}$ are in $\mathcal{N}(\mathbf{A})$. □

Example 0.4 (The solutions of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ with $\underset{\sim}{b} \neq \underset{\sim}{0}$ are not a subspace)**.** In Example 0.3, $(2,1,0)$ and $(2,0,1)$ are in $\mathcal{N}(\mathbf{A})$, and so is their sum $(4,1,1)$, since $4-2\cdot 1-2\cdot 1 = 0$. By contrast, the solutions of $\mathbf{A}\underset{\sim}{x} = (1,3)$ for the same $\mathbf{A}$ are the vectors with $x_1 - 2x_2 - 2x_3 = 1$. That set does not contain $\underset{\sim}{0}_3$, so by Theorem 1 in Subspaces and Rank it is not a subspace.

Exercise 0.6 (Showing that the solutions of two equations are a subspace)**.** Theorem 0.2 turns a question about a subspace into a question about a null space. Let $\mathcal{S}$ hold the solutions $\underset{\sim}{x} \in \mathbb{R}^3$ of the two equations $x_1 + x_2 - x_3 = 0$ and $x_2 = 2x_3$.
Show that $\mathcal{S}$ is a subspace of $\mathbb{R}^3$.

Hint

Write both equations as $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_2$ for one 2×3 matrix $\mathbf{A}$.

Solution

Solution 0.6. Move every term to the left side: $x_1 + x_2 - x_3 = 0$ and $x_2 - 2x_3 = 0$. With

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \end{bmatrix},$$

the two equations say $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_2$, so $\mathcal{S} = \mathcal{N}(\mathbf{A})$. By Theorem 0.2, $\mathcal{S}$ is a subspace of $\mathbb{R}^3$.

Theorem 0.3 (Independent columns exactly when the null space is trivial)**.** *For an $m \times n$ matrix $\mathbf{A}$, the following are equivalent:*

1. *The columns of $\mathbf{A}$ are linearly independent (Definition 1 in Subspaces and Rank).*
2. *$\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ only for $\underset{\sim}{x} = \underset{\sim}{0}_n$.*
3. *$\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$ (Definition 0.3).*

So $\mathbf{A}$ has full column rank (Definition 3 in Subspaces and Rank) exactly when its null space contains only the zero vector.

Proof

Proof. Write $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ for the columns of $\mathbf{A}$. For $\underset{\sim}{x} = (x_1, \ldots, x_n)$, the matrix-vector product is the combination of the columns with weights $x_1, \ldots, x_n$:

$$\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n.$$

So $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ says exactly that $x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n = \underset{\sim}{0}_m$. By Definition 1 in Subspaces

and Rank, the columns are linearly independent exactly when the only weights with this property are $x_1 = \cdots = x_n = 0$. That is statement 2, so statements 1 and 2 are equivalent. Statement 2 says that the only vector in $\mathcal{N}(\mathbf{A})$ is $\underset{\sim}{0}_n$, which is statement 3. The last sentence is the definition of full column rank (Definition 3 in Subspaces and Rank). □

Example 0.5 (Checking independence through the null space)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} x_1 + x_2 \\ x_1 + 2x_2 \\ x_1 + 3x_2 \end{bmatrix}.$$

This is $\underset{\sim}{0}_3$ only if $x_2 = 0$ (subtract the first entry from the second), and then $x_1 = 0$. So $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_2\}$ and the two columns are linearly independent.

For $\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$, $\mathbf{B}(2, -1) = \underset{\sim}{0}_3$, so $\mathcal{N}(\mathbf{B})$ contains a nonzero vector and the columns of $\mathbf{B}$ are linearly dependent.

Exercise 0.7 (Testing independent columns through the null space)**.** Theorem 0.3 turns a question about independence into one about solving $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{bmatrix}.$$

Decide whether the columns of $\mathbf{A}$ are linearly independent.

Hint

Solve $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_3$, starting with the third entry.

Solution

Solution 0.7. For $\underset{\sim}{x} = (x_1, x_2)$,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \\ x_2 \end{bmatrix}.$$

The third entry gives $x_2 = 0$, and then the first gives $x_1 = 0$. So $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_2\}$, and by Theorem 0.3 the columns of $\mathbf{A}$ are linearly independent.

Theorem 0.4 (The row space and null space share only the zero vector)**.** *For any $m \times n$ matrix* $\mathbf{A}$*,*

$$\mathcal{C}(\mathbf{A}^\top) \cap \mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}.$$

Proof

Proof. Both sets are subspaces, by Theorem 0.1 and Theorem 0.2, so both contain $\underset{\sim}{0}_n$ (Theorem 1 in Subspaces and Rank).
Now take any $\underset{\sim}{x}$ in both sets. Being in the row space means $\underset{\sim}{x} = \mathbf{A}^\top \underset{\sim}{u}$ for some $\underset{\sim}{u} \in \mathbb{R}^m$ (Definition 0.1), and being in the null space means $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ (Definition 0.3). Then

$$\begin{aligned}
\underset{\sim}{x}^\top \underset{\sim}{x} &= (\mathbf{A}^\top \underset{\sim}{u})^\top \underset{\sim}{x} && \text{(substitute } \underset{\sim}{x} = \mathbf{A}^\top \underset{\sim}{u} \text{ in the first factor)} \\
&= \left(\underset{\sim}{u}^\top (\mathbf{A}^\top)^\top\right) \underset{\sim}{x} && \text{(Theorem 19 in Matrices)} \\
&= (\underset{\sim}{u}^\top \mathbf{A})\, \underset{\sim}{x} && \text{(transposing twice returns the original matrix)} \\
&= \underset{\sim}{u}^\top (\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 7 in Matrices)} \\
&= \underset{\sim}{u}^\top \underset{\sim}{0}_m && \text{(substitute } \mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m) \\
&= 0. && \text{(every term of the inner product is 0)}
\end{aligned}$$

But $\underset{\sim}{x}^\top \underset{\sim}{x} = x_1^2 + \cdots + x_n^2$, which is 0 only when every $x_i = 0$. So $\underset{\sim}{x} = \underset{\sim}{0}_n$. □

Example 0.6 (The row space and null space in Example 0.3)**.** For $\mathbf{A}$ in Example 0.1, the row space is $\text{span}\{(1, -2, -2)\}$ and the null space is $\text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). A vector $c\,(1, -2, -2)$ of the row space is in the null space only if

$$\begin{aligned}
\mathbf{A}\,(c\,(1, -2, -2)) &= c\,(1 - 2 \cdot (-2) - 2 \cdot (-2),\ 3\,(1 - 2 \cdot (-2) - 2 \cdot (-2))) && \text{(Example 0.3's formula for } \mathbf{A}\underset{\sim}{x}) \\
&= c\,(9, 27) && \text{(arithmetic)}
\end{aligned}$$

is $\underset{\sim}{0}_2$, that is, only if $c = 0$. So the two subspaces share only $\underset{\sim}{0}_3$.

Exercise 0.8 (Testing a row-space vector against the null space)**.** Theorem 0.4 says that a nonzero vector in the row space is never in the null space. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \underset{\sim}{v} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix},$$

so that $\underset{\sim}{v} = (1, 1, 0) + (0, 1, 1)$ adds the two rows of $\mathbf{A}$.
Check that $\underset{\sim}{v}$ is not in $\mathcal{N}(\mathbf{A})$.

Hint

Multiply each row of $\mathbf{A}$ by $\underset{\sim}{v}$.

Solution

Solution 0.8. Here

$$\mathbf{A}\underset{\sim}{v} = \begin{bmatrix} 1+2+0 \\ 0+2+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \neq \underset{\sim}{0}_2,$$

so $\underset{\sim}{v}$ is not in $\mathcal{N}(\mathbf{A})$, as Theorem 0.4 requires of a nonzero vector in the row space.

Definition 0.5 (Cross-product matrix)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with columns $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ and $\mathbf{B}$ an $m \times q$ matrix with columns $\underset{\sim}{b}_1, \ldots, \underset{\sim}{b}_q$. The **cross-product matrix** of $\mathbf{A}$ and $\mathbf{B}$ is the $n \times q$ matrix $\mathbf{A}^\top\mathbf{B}$, whose entry (i, j) is the dot product $\underset{\sim}{a}_i \cdot \underset{\sim}{b}_j$.

This is what R's `crossprod(A, B)` computes, and `crossprod(A)` is $\mathbf{A}^\top\mathbf{A}$. In NumPy it has no name of its own: write `A.T @ B`.

For two column vectors it is the 1×1 matrix holding their dot product. It is unrelated to the cross product $\underset{\sim}{u} \times \underset{\sim}{v}$ of two vectors in $\mathbb{R}^3$, and differs from the Frobenius inner product $\operatorname{tr}(\mathbf{A}^\top\mathbf{B})$, which sums the diagonal of this matrix into a single number.

Example 0.7 (A cross-product matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Then $\mathbf{A}^\top\mathbf{B}$ is 2×1, and

$$\begin{aligned} \mathbf{A}^\top\mathbf{B} &= \begin{bmatrix} \underset{\sim}{a}_1 \cdot \underset{\sim}{b}_1 \\ \underset{\sim}{a}_2 \cdot \underset{\sim}{b}_1 \end{bmatrix} && \text{(Definition 0.5)} \\ &= \begin{bmatrix} 1+2+0 \\ 0+2+3 \end{bmatrix} && \text{(multiply and add)} \\ &= \begin{bmatrix} 3 \\ 5 \end{bmatrix}. && \text{(add)} \end{aligned}$$

Exercise 0.9 (Computing a cross-product matrix)**.** In R, `crossprod(A, B)` computes the cross-product matrix $\mathbf{A}^\top\mathbf{B}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Compute $\mathbf{A}^\top \mathbf{B}$.

Hint

Entry i is the dot product of column i of $\mathbf{A}$ with $(2, 3)$.

Solution

Solution 0.9. The columns of $\mathbf{A}$ are $(1, 1)$ and $(0, 1)$, so

$$\mathbf{A}^\top \mathbf{B} = \begin{bmatrix} 1 \cdot 2 + 1 \cdot 3 \\ 0 \cdot 2 + 1 \cdot 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

Remark (Column-column products)

Remark 0.1 (Column-column products). Ordinary matrix multiplication pairs a row of the left factor with a column of the right factor, which makes each entry a row-by-column product [2, Sec. 2]. Each entry of a cross-product matrix instead pairs a column of $\mathbf{A}$ with a column of $\mathbf{B}$, so in these notes we call it a column-column product.

Definition 0.6 (Gram matrix)**.** The **Gram matrix** [3] of an $m \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\mathbf{A}^\top \mathbf{A}$, the cross-product matrix of $\mathbf{A}$ with itself.

Example 0.8 (A Gram matrix and its entries)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, with columns $\underset{\sim}{a}_1 = (1, 1, 0)$ and $\underset{\sim}{a}_2 = (0, 1, 1)$. Then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} && \text{(Definition 6 in Matrices)} \\ &= \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix} && \text{(Definition 10 in Matrices)} \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. && \text{(multiply and add)} \end{aligned}$$

Row i of $\mathbf{A}^\top$ is column i of $\mathbf{A}$, so entry (i, j) of the Gram matrix is the dot product $\underset{\sim}{a}_i \cdot \underset{\sim}{a}_j$: for instance,

$$\begin{aligned} \underset{\sim}{a}_1 \cdot \underset{\sim}{a}_2 &= 0 + 1 + 0 \\ &= 1 \end{aligned}$$

is entry $(1, 2)$.

Exercise 0.10 (Computing a Gram matrix)**.** The Gram matrix holds the dot products of every pair of columns of $\mathbf{A}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute $\mathbf{A}^\top \mathbf{A}$.

Hint

Entry (i, j) is $\underset{\sim}{a}_i \cdot \underset{\sim}{a}_j$, so entries $(1, 2)$ and $(2, 1)$ agree.

Solution

Solution 0.10. The columns of $\mathbf{A}$ are $\underset{\sim}{a}_1 = (1, 0, 1)$ and $\underset{\sim}{a}_2 = (2, 1, 0)$, so

$$\begin{aligned} \underset{\sim}{a}_1 \cdot \underset{\sim}{a}_1 &= 1 + 0 + 1 = 2, \\ \underset{\sim}{a}_1 \cdot \underset{\sim}{a}_2 &= 2 + 0 + 0 = 2, \\ \underset{\sim}{a}_2 \cdot \underset{\sim}{a}_2 &= 4 + 1 + 0 = 5, \end{aligned}$$

and

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}.$$

Theorem 0.5 (The cross-product matrix is the sum of the outer products of the rows)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with rows* $\underset{\sim}{r}_1^\top, \ldots, \underset{\sim}{r}_m^\top$*. Let* $\mathbf{B}$ *be an* $m \times q$ *matrix with rows* $\underset{\sim}{s}_1^\top, \ldots, \underset{\sim}{s}_m^\top$*. Then the cross-product matrix of* $\mathbf{A}$ *with* $\mathbf{B}$ *is*

$$\mathbf{A}^\top \mathbf{B} = \sum_{i=1}^{m} \underset{\sim}{r}_i \, \underset{\sim}{s}_i^\top.$$

Proof

Proof. Compare entry (j,k) of each side.

$$\begin{aligned}
(\mathbf{A}^\top\mathbf{B})_{jk} &= \sum_{i=1}^{m}(\mathbf{A})_{ij}(\mathbf{B})_{ik} && \text{(Definition 10 in Matrices)}\\
&= \sum_{i=1}^{m}(\underset{\sim}{r}_i)_j(\underset{\sim}{s}_i)_k && \text{(entries of the rows)}\\
&= \sum_{i=1}^{m}(\underset{\sim}{r}_i\,\underset{\sim}{s}_i{}^\top)_{jk} && \text{(Definition 16 in Inner Products and Orthogonality)}\\
&= \left(\sum_{i=1}^{m}\underset{\sim}{r}_i\,\underset{\sim}{s}_i{}^\top\right)_{jk}. && \text{(entries of a sum)}
\end{aligned}$$

Every entry agrees, so the two matrices agree. □

Example 0.9 (The Gram matrix and a cross-product as sums of outer products)**.** Let $\mathbf{A} = \begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}$, $\mathbf{B} = \begin{bmatrix}1\\2\\3\end{bmatrix}$ be the matrices of Example 0.7. The rows of $\mathbf{A}$ are $\underset{\sim}{r}_1 = (1,0)$, $\underset{\sim}{r}_2 = (1,1)$, $\underset{\sim}{r}_3 = (0,1)$. Taking $\mathbf{B} = \mathbf{A}$ in Theorem 0.5, the Gram matrix is the sum of the outer products of the rows with themselves:

$$\begin{aligned}
\mathbf{A}^\top\mathbf{A} &= \underset{\sim}{r}_1\,\underset{\sim}{r}_1{}^\top + \underset{\sim}{r}_2\,\underset{\sim}{r}_2{}^\top + \underset{\sim}{r}_3\,\underset{\sim}{r}_3{}^\top && \text{(Theorem 0.5)}\\
&= \begin{bmatrix}1&0\\0&0\end{bmatrix} + \begin{bmatrix}1&1\\1&1\end{bmatrix} + \begin{bmatrix}0&0\\0&1\end{bmatrix} && \text{(Definition 16 in Inner Products and Orthogonality)}\\
&= \begin{bmatrix}2&1\\1&2\end{bmatrix}. && \text{(add)}
\end{aligned}$$

This is the Gram matrix of Example 0.8. Now keep the column $\mathbf{B}$ above. Its rows are the 1-vectors $\underset{\sim}{s}_1 = (1)$, $\underset{\sim}{s}_2 = (2)$, $\underset{\sim}{s}_3 = (3)$, so

$$\begin{aligned}
\mathbf{A}^\top\mathbf{B} &= 1\underset{\sim}{r}_1 + 2\underset{\sim}{r}_2 + 3\underset{\sim}{r}_3 && \text{(Theorem 0.5)}\\
&= \begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}2\\2\end{bmatrix} + \begin{bmatrix}0\\3\end{bmatrix} && \text{(scalar multiples)}\\
&= \begin{bmatrix}3\\5\end{bmatrix}, && \text{(add)}
\end{aligned}$$

the same as in Example 0.7.

Exercise 0.11 (Computing a cross-product matrix one row at a time)**.** Theorem 0.5 builds a cross-product matrix by adding one outer product per row, which is how it can be updated as observations arrive. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Compute $\mathbf{A}^\top \mathbf{B}$ by adding two outer products.

Hint

Rows of $\mathbf{A}$ are $(1, 2)$ and $(0, 1)$, and rows of $\mathbf{B}$ are 3 and 1.

Solution

Solution 0.11. By Theorem 0.5,

$$\mathbf{A}^\top \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3] + \begin{bmatrix} 0 \\ 1 \end{bmatrix} [1] = \begin{bmatrix} 3 \\ 6 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}.$$

Multiplying directly gives the same: $\mathbf{A}^\top \mathbf{B} = (1 \cdot 3 + 0 \cdot 1,\ 2 \cdot 3 + 1 \cdot 1) = (3, 7)$.

Theorem 0.6 ($\mathbf{A}^\top \mathbf{A}$ has the same null space as $\mathbf{A}$)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}).$$

Proof

Proof. $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top \mathbf{A})$**.** If $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$, then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A}\underset{\sim}{x} &= \mathbf{A}^\top \underset{\sim}{0}_m \\ &= \underset{\sim}{0}_n. \end{aligned}$$

$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$**.** If $\mathbf{A}^\top \mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_n$, then

$$\begin{aligned} 0 &= \underset{\sim}{x}^\top \underset{\sim}{0}_n && \text{(every term of the inner product is 0)} \\ &= \underset{\sim}{x}^\top (\mathbf{A}^\top \mathbf{A}\underset{\sim}{x}) && \text{(substitute } \underset{\sim}{0}_n = \mathbf{A}^\top \mathbf{A}\underset{\sim}{x}) \\ &= (\underset{\sim}{x}^\top \mathbf{A}^\top)(\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 7 in Matrices)} \\ &= (\mathbf{A}\underset{\sim}{x})^\top (\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 19 in Matrices)} \\ &= (\mathbf{A}\underset{\sim}{x}) \cdot (\mathbf{A}\underset{\sim}{x}) && \text{(Example 8 in Matrices)} \\ &= \|\mathbf{A}\underset{\sim}{x}\|^2. && \text{(square both sides of Equation 2 in Vectors)} \end{aligned}$$

A vector whose Euclidean norm is 0 has every entry 0, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$. □

Example 0.10 (The null space of a Gram matrix)**.** For $\mathbf{A}$ in Example 0.1,

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix} = \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}.$$

Every row is a multiple of $(1, -2, -2)$, so $\mathbf{A}^\top \mathbf{A} \underset{\sim}{x} = \underset{\sim}{0}_3$ exactly when $x_1 - 2x_2 - 2x_3 = 0$, the same condition as for $\mathcal{N}(\mathbf{A})$ in Example 0.3.

Exercise 0.12 (Finding the null space of a Gram matrix)**.** Theorem 0.6 lets the null space of a Gram matrix be read off from the original matrix. Let $\mathbf{A}$ be the 3×2 matrix whose every entry is 1.
Find $\mathcal{N}(\mathbf{A}^\top \mathbf{A})$.

Hint

Find $\mathcal{N}(\mathbf{A})$ first. Each entry of $\mathbf{A}\underset{\sim}{x}$ is $x_1 + x_2$.

Solution

Solution 0.12. Each entry of $\mathbf{A}\underset{\sim}{x}$ is $x_1 + x_2$, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_3$ exactly when $x_2 = -x_1$, and $\mathcal{N}(\mathbf{A}) = \text{span}\{(1, -1)\}$. By Theorem 0.6,

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \text{span}\{(1, -1)\}.$$

As a check, $\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$, which sends $(1, -1)$ to $\underset{\sim}{0}_2$.

Rank-nullity and rank factorization

Source

This section is adapted from Zhou [4], used under the MIT License (see the license text in Section in Subspaces and Rank). Theorem 0.5 and Corollary 0.2 restore the source's statements that a product has rank at most the rank of either factor, and that multiplying by a matrix of full column rank or by a matrix whose rank is its number of rows does not change the rank. These parts of the source are not part of this section:

- the names "full row rank", "full rank" and "singular"
- the statement that $\mathcal{C}(\mathbf{C}) \subseteq \mathcal{C}(\mathbf{A})$ exactly when $\mathbf{C} = \mathbf{AB}$ for some $\mathbf{B}$, and its row space version, and the containment $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$
- the dimension of a subspace as the largest number of linearly independent vectors in it, or the smallest number of vectors that span it
- the properties of a rank factorization $\mathbf{A} = \mathbf{CR}$ ($\text{rank}(\mathbf{C}) = \text{rank}(\mathbf{R}) = r$, $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{C})$, $\mathcal{N}(\mathbf{A}) = \mathcal{N}(\mathbf{R})$), the fact that it is not unique, and the second proof of $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$ that it gives
- calling $\mathbf{AA}^\top$ a Gram matrix, and the equation $\mathcal{N}(\mathbf{AA}^\top) = \mathcal{N}(\mathbf{A}^\top)$ (Theorem 0.6 applied to $\mathbf{A}^\top$ gives it)
- the statement that $\mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{A}^\top)$ share only the zero vector (it follows from Theorem 10 in Direct Sums and Orthogonal Complements)

Theorem 0.1 (The rank is the dimension of the column space)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\text{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A})).$$

Proof

Proof. Let $r = \text{rank}(\mathbf{A})$ (Definition 2 in Subspaces and Rank), and choose r linearly independent columns of $\mathbf{A}$; by Definition 2 in Subspaces and Rank, no larger set of columns is linearly independent. We show the chosen columns are a basis of $\mathcal{C}(\mathbf{A})$ (Definition 6 in Subspaces and Rank).

Every column is in the span of the chosen columns. A chosen column is in that span trivially. If some unchosen column $\underset{\sim}{a}_j$ were not in the span, then adding it to the chosen columns would give $r+1$ linearly independent columns (Theorem 8 in Subspaces and Rank), more than r. So every column of $\mathbf{A}$ is in the span of the chosen columns.

The chosen columns span $\mathcal{C}(\mathbf{A})$. The span of the chosen columns is a subspace (Theorem 2 in Subspaces and Rank), so it is closed under addition and scalar multiplication, and it contains every linear combination of the columns of $\mathbf{A}$. By Theorem 0.1, those combinations make up $\mathcal{C}(\mathbf{A})$, so $\mathcal{C}(\mathbf{A})$ is contained in the span of the chosen columns. The chosen columns are themselves columns of $\mathbf{A}$, so their span is contained in $\mathcal{C}(\mathbf{A})$, and the two sets are equal.

The chosen columns are linearly independent and span $\mathcal{C}(\mathbf{A})$, so they are a basis of it, and $\dim(\mathcal{C}(\mathbf{A})) = r$ (Definition 9 in Subspaces and Rank). If $r = 0$, every column is $\underset{\sim}{0}_m$, $\mathcal{C}(\mathbf{A}) = \{\underset{\sim}{0}_m\}$, and both sides are 0. □

Example 0.1 (The rank of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1,3)\}$ (Example 0.2), and $(1,3)$ is linearly independent because it is nonzero, so it

is a basis and $\mathcal{C}(\mathbf{A})$ has dimension 1. So $\text{rank}(\mathbf{A}) = 1$, by Theorem 0.1. Directly: the first column $(1, 3)$ is linearly independent on its own, and any two columns are dependent, because each column is a multiple of $(1, 3)$.

Exercise 0.1 (Finding a rank from the column space)**.** Theorem 0.1 reads the rank of a matrix off the dimension of its column space. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

Find $\text{rank}(\mathbf{A})$ by finding $\dim(\mathcal{C}(\mathbf{A}))$.

Hint

In $\mathbf{A}$, the second column is a multiple of the first.

Solution

Solution 0.1. In $\mathbf{A}$, the second column is $(2, 4, 6) = 2\,(1, 2, 3)$, so by Theorem 0.1

$$\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 2, 3), (2, 4, 6)\} = \text{span}\{(1, 2, 3)\}.$$

The single nonzero vector $(1, 2, 3)$ is a basis of that span, so $\dim(\mathcal{C}(\mathbf{A})) = 1$, and by Theorem 0.1, $\text{rank}(\mathbf{A}) = 1$.

Definition 0.1 (Nullity)**.** The **nullity** [5] of a matrix $\mathbf{A}$ is the dimension of its null space (Definition 0.3):

$$\text{nullity}(\mathbf{A}) \stackrel{\text{def}}{=} \dim(\mathcal{N}(\mathbf{A})).$$

Example 0.2 (The nullity of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). Those two vectors are linearly independent: $c_1(2, 1, 0) + c_2(2, 0, 1) = (2c_1 + 2c_2, c_1, c_2)$, which is $\underset{\sim}{0}_3$ only if

$$\begin{aligned} c_1 &= c_2 \\ &= 0. \end{aligned}$$

So they are a basis of $\mathcal{N}(\mathbf{A})$, and $\text{nullity}(\mathbf{A}) = 2$. The 2×2 identity matrix has nullity 0, because $\mathbf{I}\underset{\sim}{x} = \underset{\sim}{x}$ is $\underset{\sim}{0}_2$ only when $\underset{\sim}{x} = \underset{\sim}{0}_2$.

Theorem 0.2 (Rank-nullity theorem)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = n.$$

Proof

Proof. Let $\nu = \operatorname{nullity}(\mathbf{A})$. Choose a basis $\underset{\sim}{x}_1, \dots, \underset{\sim}{x}_\nu$ of $\mathcal{N}(\mathbf{A})$, and extend it to a basis $\underset{\sim}{x}_1, \dots, \underset{\sim}{x}_\nu, \underset{\sim}{y}_1, \dots, \underset{\sim}{y}_{n-\nu}$ of $\mathbb{R}^n$ (Theorem 9 in Subspaces and Rank); it has n vectors, because every basis of $\mathbb{R}^n$ has $\dim(\mathbb{R}^n) = n$ vectors (Theorem 7 in Subspaces and Rank, Example 16 in Subspaces and Rank). We show that $\mathbf{A}\underset{\sim}{y}_1, \dots, \mathbf{A}\underset{\sim}{y}_{n-\nu}$ are a basis of $\mathcal{C}(\mathbf{A})$, using Theorem 4 in Subspaces and Rank to apply $\mathbf{A}$ to linear combinations.

They are linearly independent. Suppose $\sum_{i=1}^{n-\nu} v_i \mathbf{A}\underset{\sim}{y}_i = \underset{\sim}{0}_m$. Then $\mathbf{A}\left(\sum_i v_i \underset{\sim}{y}_i\right) = \underset{\sim}{0}_m$ (Theorem 4 in Subspaces and Rank), so $\sum_i v_i \underset{\sim}{y}_i \in \mathcal{N}(\mathbf{A})$, and it equals $\sum_{j=1}^{\nu} u_j \underset{\sim}{x}_j$ for some numbers u_j, since the $\underset{\sim}{x}_j$ span $\mathcal{N}(\mathbf{A})$. Moving everything to one side,

$$\sum_{j=1}^{\nu} (-u_j)\, \underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} v_i \underset{\sim}{y}_i = \underset{\sim}{0}_n,$$

and because the $\underset{\sim}{x}$'s and $\underset{\sim}{y}$'s together form a basis of $\mathbb{R}^n$, which is linearly independent, every coefficient is 0; in particular every $v_i = 0$.

They span $\mathcal{C}(\mathbf{A})$. Take any $\underset{\sim}{w} \in \mathcal{C}(\mathbf{A})$, so $\underset{\sim}{w} = \mathbf{A}\underset{\sim}{z}$ for some $\underset{\sim}{z} \in \mathbb{R}^n$ (Definition 0.1). Write $\underset{\sim}{z} = \sum_j a_j \underset{\sim}{x}_j + \sum_i b_i \underset{\sim}{y}_i$ in the basis of $\mathbb{R}^n$. Then

$$\begin{aligned}
\underset{\sim}{w} &= \mathbf{A}\left(\sum_{j=1}^{\nu} a_j \underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} b_i \underset{\sim}{y}_i\right) && \text{(substitute } \underset{\sim}{z}\text{)} \\
&= \sum_{j=1}^{\nu} a_j\, \mathbf{A}\underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} b_i\, \mathbf{A}\underset{\sim}{y}_i && \text{(Theorem 4 in Subspaces and Rank)} \\
&= \sum_{j=1}^{\nu} a_j\, \underset{\sim}{0}_m + \sum_{i=1}^{n-\nu} b_i\, \mathbf{A}\underset{\sim}{y}_i && \text{(each } \underset{\sim}{x}_j \in \mathcal{N}(\mathbf{A})\text{)} \\
&= \sum_{i=1}^{n-\nu} b_i\, \mathbf{A}\underset{\sim}{y}_i, && \text{(drop the zero terms)}
\end{aligned}$$

a linear combination of $\mathbf{A}\underset{\sim}{y}_1, \dots, \mathbf{A}\underset{\sim}{y}_{n-\nu}$. These vectors are in $\mathcal{C}(\mathbf{A})$, so their span is exactly $\mathcal{C}(\mathbf{A})$.

So $\dim(\mathcal{C}(\mathbf{A})) = n - \nu$, and $\operatorname{rank}(\mathbf{A}) = n - \nu$ by Theorem 0.1. □

Example 0.3 (Checking rank-nullity on the matrix in Example 0.1)**.** The matrix $\mathbf{A}$ in Example 0.1 has $n = 3$ columns, rank 1 (Example 0.1) and nullity 2 (Example 0.2), and $1+2 = 3$. The 2×2 identity matrix has rank 2 and nullity 0 (Example 0.2), and $2+0 = 2$.

Exercise 0.2 (Finding the nullity of a 3×3 matrix)**.** The nullity of a matrix is the dimension of its null space. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Find nullity($\mathbf{A}$).

Hint

Write out $\mathbf{A}\underset{\sim}{x}$, find the null space, and then find a basis of it.

Solution

Solution 0.2. For $\underset{\sim}{x} = (x_1, x_2, x_3)$, $\mathbf{A}\underset{\sim}{x} = (2x_1, 0, 5x_3)$, which is $\underset{\sim}{0}_3$ exactly when $x_1 = 0$ and $x_3 = 0$. So $\mathcal{N}(\mathbf{A}) = \{(0, x_2, 0) : x_2 \in \mathbb{R}\} = \text{span}\{\underset{\sim}{e}_2\}$. The single nonzero vector $\underset{\sim}{e}_2$ is a basis of it, so nullity($\mathbf{A}$) $= 1$.

Exercise 0.3 (Finding a nullity from a rank)**.** Suppose that a 5×8 matrix $\mathbf{A}$ has rank 3.
Find nullity($\mathbf{A}$).

Hint

Use Theorem 0.2 with $n = 8$.

Solution

Solution 0.3. By Theorem 0.2, rank($\mathbf{A}$) + nullity($\mathbf{A}$) $= 8$, so

$$\text{nullity}(\mathbf{A}) = 8 - 3 = 5.$$

Theorem 0.3 (Multiplying on the right cannot increase the rank)**.** *If* $\mathbf{A}$ *is* $m \times n$ *and* $\mathbf{B}$ *is* $n \times k$*, then*

$$\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{A}).$$

Proof

Proof. $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$**.** Any vector in $\mathcal{C}(\mathbf{AB})$ is $(\mathbf{AB})\,\underset{\sim}{x}$ for some $\underset{\sim}{x} \in \mathbb{R}^k$ (Definition 0.1), and

$$(\mathbf{AB})\,\underset{\sim}{x} = \mathbf{A}\,(\mathbf{B}\underset{\sim}{x}) \quad \text{(Theorem 7 in Matrices)}$$

is $\mathbf{A}$ times the vector $\mathbf{B}\underset{\sim}{x} \in \mathbb{R}^n$, so it is in $\mathcal{C}(\mathbf{A})$.
Compare dimensions. Both column spaces are subspaces (Theorem 0.1), so

$$\begin{aligned} \operatorname{rank}(\mathbf{AB}) &= \dim(\mathcal{C}(\mathbf{AB})) && \text{(Theorem 0.1)} \\ &\leq \dim(\mathcal{C}(\mathbf{A})) && \text{(Theorem 10 in Subspaces and Rank, part 2)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.1)} \end{aligned}$$

□

Example 0.4 (A product can lose rank)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{AB} = \begin{bmatrix} 1\cdot 0 + 0\cdot 1 & 1\cdot 0 + 0\cdot 0 \\ 0\cdot 0 + 0\cdot 1 & 0\cdot 0 + 0\cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 1, but $\mathbf{AB}$ has rank 0, so the inequality in Theorem 0.3 can be strict. With the 2×2 identity in place of $\mathbf{B}$, $\mathbf{AI} = \mathbf{A}$ keeps rank 1, and the inequality is an equality.

Exercise 0.4 (Finding how large the rank of a matrix product can be)**.** Theorem 0.3 shows how large the rank of a matrix product can be. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}.$$

Find the number that, by Theorem 0.3, $\operatorname{rank}(\mathbf{AB})$ cannot exceed.

Hint

Find $\operatorname{rank}(\mathbf{A})$ first.

Solution

Solution 0.4. Columns of $\mathbf{A}$ are $(1, 0)$ and $\underset{\sim}{0}_2$, so its column space is $\operatorname{span}\{(1, 0)\}$, and Theorem 0.1 gives $\operatorname{rank}(\mathbf{A}) = 1$. By Theorem 0.3, $\operatorname{rank}(\mathbf{AB}) \leq 1$. Indeed, $\mathbf{AB} = \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix}$, which has rank 1.

Theorem 0.4 (Fundamental theorem of ranks)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\begin{aligned}\operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top) \\ &= \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top).\end{aligned}$$

In particular, the largest number of linearly independent rows of $\mathbf{A}$ equals the largest number of linearly independent columns.

Proof

Proof. $\operatorname{rank}(\mathbf{A}^\top\mathbf{A}) = \operatorname{rank}(\mathbf{A})$**.** Both matrices have n columns, and $\mathcal{N}(\mathbf{A}^\top\mathbf{A}) = \mathcal{N}(\mathbf{A})$ (Theorem 0.6), so they have the same nullity (Definition 0.1). Then

$$\begin{aligned}\operatorname{rank}(\mathbf{A}^\top\mathbf{A}) &= n - \operatorname{nullity}(\mathbf{A}^\top\mathbf{A}) && \text{(Theorem 0.2)} \\ &= n - \operatorname{nullity}(\mathbf{A}) && \text{(equal null spaces)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.2)}\end{aligned}$$

$\operatorname{rank}(\mathbf{A}\mathbf{A}^\top) = \operatorname{rank}(\mathbf{A}^\top)$**.** Apply the previous step to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$.

Chain the inequalities.

$$\begin{aligned}\operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) && \text{(first step)} \\ &\le \operatorname{rank}(\mathbf{A}^\top) && \text{(Theorem 0.3, with } \mathbf{A}^\top \text{ on the left)} \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top) && \text{(second step)} \\ &\le \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.3, with } \mathbf{A} \text{ on the left)}\end{aligned}$$

The chain starts and ends at $\operatorname{rank}(\mathbf{A})$, so every quantity in it equals $\operatorname{rank}(\mathbf{A})$. The columns of $\mathbf{A}^\top$ are the rows of $\mathbf{A}$, which gives the statement about rows. □

Example 0.5 (Four equal ranks)**.** For $\mathbf{A}$ in Example 0.1, $\operatorname{rank}(\mathbf{A}) = 1$ (Example 0.1). Its transpose $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}$ has second column 3 times its first, so its rank is 1. $\mathbf{A}^\top\mathbf{A}$ in Example 0.10 has every row a multiple of $(1, -2, -2)$, so its rank is 1 too, and

$$\begin{aligned}\mathbf{A}\mathbf{A}^\top &= \begin{bmatrix} 1\cdot 1 + (-2)(-2) + (-2)(-2) & 1\cdot 3 + (-2)(-6) + (-2)(-6) \\ 3\cdot 1 + (-6)(-2) + (-6)(-2) & 3\cdot 3 + (-6)(-6) + (-6)(-6) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 27 \\ 27 & 81 \end{bmatrix},\end{aligned}$$

whose second column is 3 times its first, so its rank is 1.

Corollary 0.1 (The rank is at most the smaller dimension)**.** *For any $m \times n$ matrix* $\mathbf{A}$*,* $\operatorname{rank}(\mathbf{A}) \leq \min\{m, n\}$.

Proof

Proof. $\mathbf{A}$ has n columns, so at most n of them are linearly independent, and $\operatorname{rank}(\mathbf{A}) \leq n$ (Definition 2 in Subspaces and Rank). Likewise $\mathbf{A}^\top$ has m columns, so $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^\top) \leq m$ (Theorem 0.4). □

Example 0.6 (A 2×3 matrix has rank at most 2)**.** By Corollary 0.1, every 2×3 matrix has rank at most $\min\{2, 3\} = 2$, even though it has 3 columns. The matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Remark 1 in Subspaces and Rank reaches that bound: its first two columns $(1, 0)$ and $(0, 1)$ are linearly independent, so its rank is 2.

Exercise 0.5 (Finding the rank of a Gram matrix without computing it)**.** A Gram matrix can be larger than the matrix it comes from, but Theorem 0.4 says it has the same rank. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}.$$

Find $\operatorname{rank}(\mathbf{A}^\top \mathbf{A})$ without computing $\mathbf{A}^\top \mathbf{A}$.

Hint

Find $\operatorname{rank}(\mathbf{A})$ from its two rows, using Theorem 0.4 once more.

Solution

Solution 0.5. Both rows of $\mathbf{A}$, $(1, 0, 2)$ and $(0, 1, 1)$, are linearly independent: $c_1(1, 0, 2) + c_2(0, 1, 1) = (c_1, c_2, 2c_1 + c_2)$ is $\underset{\sim}{0}_3$ only if $c_1 = c_2 = 0$. So $\operatorname{rank}(\mathbf{A}^\top) = 2$, and by Theorem 0.4

$$\operatorname{rank}(\mathbf{A}^\top \mathbf{A}) = \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^\top) = 2,$$

even though $\mathbf{A}^\top \mathbf{A}$ is 3×3.

Exercise 0.6 (Finding how large the rank of a tall matrix can be)**.** A table of 5 observations of 3 features is a tall 5×3 matrix. Corollary 0.1 says how large its rank can be.

Find the largest rank that a 5×3 matrix can have.

Hint

Use Corollary 0.1 with $m = 5$ and $n = 3$.

Solution

Solution 0.6. By Corollary 0.1, the rank is at most $\min\{5, 3\} = 3$. A rank of 3 is reached, for example by the 5×3 matrix with rows $\underset{\sim}{e}_1{}^\top, \underset{\sim}{e}_2{}^\top, \underset{\sim}{e}_3{}^\top, \underset{\sim}{0}_3{}^\top, \underset{\sim}{0}_3{}^\top$, whose three columns are linearly independent.

Theorem 0.5 (Multiplying on the left cannot increase the rank)**.** *If* $\mathbf{A}$ *is* $m \times n$ *and* $\mathbf{B}$ *is* $n \times k$*, then*

$$\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}(\mathbf{B}).$$

Proof

Proof.

$$\begin{aligned}
\operatorname{rank}(\mathbf{AB}) &= \operatorname{rank}((\mathbf{AB})^\top) && \text{(Theorem 0.4)} \\
&= \operatorname{rank}(\mathbf{B}^\top \mathbf{A}^\top) && \text{(Theorem 19 in Matrices)} \\
&\leq \operatorname{rank}(\mathbf{B}^\top) && \text{(Theorem 0.3, with } \mathbf{B}^\top \text{ on the left)} \\
&= \operatorname{rank}(\mathbf{B}). && \text{(Theorem 0.4)}
\end{aligned}$$

□

Example 0.7 (A product of rank 1 with a factor of rank 2)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{AB} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 2, because its columns $(1, 0)$ and $(1, 1)$ are linearly independent. $\mathbf{B}$ has rank 1, because its second column is $\underset{\sim}{0}_2$ and its first is not, and $\mathbf{AB}$ has rank 1 for the same reason. So $\operatorname{rank}(\mathbf{AB}) = 1 \leq \operatorname{rank}(\mathbf{B}) = 1$ (Theorem 0.5), and the bound of Theorem 0.3, 2, is not the tighter one.

Exercise 0.7 (Finding how large the rank of a product can be from its right factor)**.** Theorem 0.5 shows how large the rank of a matrix product can be, given the rank of its right factor. Let $\mathbf{A}$ be any 3×2 matrix, and let

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Find the number that, by Theorem 0.5, $\operatorname{rank}(\mathbf{AB})$ cannot exceed, whatever $\mathbf{A}$ is.

Hint

Find $\operatorname{rank}(\mathbf{B})$ first.

Solution

Solution 0.7. Each of the two columns of $\mathbf{B}$ is $(1, 1)$, so its column space is $\operatorname{span}\{(1, 1)\}$, and Theorem 0.1 gives $\operatorname{rank}(\mathbf{B}) = 1$. By Theorem 0.5, $\operatorname{rank}(\mathbf{AB}) \leq 1$ for every 3×2 matrix $\mathbf{A}$.

Corollary 0.2 (Multiplying by a matrix of full rank does not change the rank)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *and* $\mathbf{B}$ *be* $n \times k$*.*

1. *If* $\mathbf{A}$ *has full column rank (Definition 3 in Subspaces and Rank), then* $\operatorname{rank}(\mathbf{AB}) = \operatorname{rank}(\mathbf{B})$.
2. *If* $\operatorname{rank}(\mathbf{B}) = n$*, then* $\operatorname{rank}(\mathbf{AB}) = \operatorname{rank}(\mathbf{A})$.

Proof

Proof. **Part 1.** Because $\mathbf{A}$ has full column rank, $\mathbf{A}\underset{\sim}{y} = \underset{\sim}{0}_m$ only for $\underset{\sim}{y} = \underset{\sim}{0}_n$ (Theorem 0.3). So for $\underset{\sim}{x} \in \mathbb{R}^k$,

$$\mathbf{AB}\underset{\sim}{x} = \underset{\sim}{0}_m \iff \mathbf{B}\underset{\sim}{x} = \underset{\sim}{0}_n \quad \text{(Theorem 0.3, with } \underset{\sim}{y} = \mathbf{B}\underset{\sim}{x}\text{)}$$

so $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$ (Definition 0.3), and the two matrices have the same nullity (Definition 0.1). Both have k columns, so by Theorem 0.2 they have the same rank.
Part 2. $\mathcal{C}(\mathbf{B})$ is a subspace of $\mathbb{R}^n$ (Theorem 0.1) of dimension n (Theorem 0.1), so $\mathcal{C}(\mathbf{B}) = \mathbb{R}^n$ (Theorem 11 in Subspaces and Rank). Every vector of $\mathcal{C}(\mathbf{AB})$ is $\mathbf{A}\,(\mathbf{B}\underset{\sim}{x})$ for some $\underset{\sim}{x}$ (Definition 0.1), and $\mathbf{B}\underset{\sim}{x}$ takes every vector of $\mathbb{R}^n$, so $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$ and the two matrices have the same rank (Theorem 0.1). □

Example 0.8 (Ranks of products with a full-rank factor)**.** Let $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, which has rank 1.

- Part 1: $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ has two linearly independent columns, so it has full column rank, and $\mathbf{AB} = \begin{bmatrix} 2 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$ has rank $1 = \text{rank}(\mathbf{B})$.
- Part 2: let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, which has rank 1, and let $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, which has rank 2. Then $\mathbf{AB} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ has rank $1 = \text{rank}(\mathbf{A})$.

Exercise 0.8 (Finding the rank of a product with a full-column-rank factor)**.** Corollary 0.2 says that a left factor with full column rank keeps the rank of the right factor. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Find $\text{rank}(\mathbf{AB})$ without multiplying.

Hint

Check that $\mathbf{A}$ has full column rank, and find $\text{rank}(\mathbf{B})$.

Solution

Solution 0.8. Columns of $\mathbf{A}$, $(1, 0, 1)$ and $(0, 1, 1)$, are linearly independent: the first two entries of $c_1 (1, 0, 1) + c_2 (0, 1, 1)$ are c_1 and c_2. So $\mathbf{A}$ has full column rank. The second column of $\mathbf{B}$ is twice the first, so $\text{rank}(\mathbf{B}) = 1$. By part 1 of Corollary 0.2,

$$\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}) = 1.$$

Theorem 0.6 (A centered data matrix has rank at most $n-1$)**.** *Let $\mathbf{X}$ be an $n \times p$ matrix with rows $\underset{\sim}{x}_1^\top, \ldots, \underset{\sim}{x}_n^\top$, let $\bar{\underset{\sim}{x}} = \frac{1}{n} \sum_{i=1}^n \underset{\sim}{x}_i$ be the mean of the rows, and let $\mathbf{X}_c = \mathbf{X} - \underset{\sim}{1} \bar{\underset{\sim}{x}}^\top$ be the matrix whose row i is $(\underset{\sim}{x}_i - \bar{\underset{\sim}{x}})^\top$, where $\underset{\sim}{1}$ is the ones vector of length n (Definition 16 in Vectors). Then* $\text{rank}(\mathbf{X}_c) \leq n - 1$.

Proof

Proof. The entries of each column of $\mathbf{X}_c$ sum to zero:

$$\begin{aligned}
\underset{\sim}{1}^\top \mathbf{X}_c &= \underset{\sim}{1}^\top \mathbf{X} - (\underset{\sim}{1}^\top \underset{\sim}{1})\bar{\underset{\sim}{x}}^\top && \text{(Theorem 8 in Matrices)} \\
&= \sum_{i=1}^{n} \underset{\sim}{x_i}^\top - n\,\bar{\underset{\sim}{x}}^\top && (\underset{\sim}{1}^\top \underset{\sim}{1} = n) \\
&= n\,\bar{\underset{\sim}{x}}^\top - n\,\bar{\underset{\sim}{x}}^\top && \text{(definition of } \bar{\underset{\sim}{x}}) \\
&= \underset{\sim}{0_p}^\top.
\end{aligned}$$

Transposing, $\mathbf{X}_c{}^\top \underset{\sim}{1} = \underset{\sim}{0}_p$. The ones vector is nonzero, so it is in the null space of the $p \times n$ matrix $\mathbf{X}_c{}^\top$, and $\text{nullity}(\mathbf{X}_c{}^\top) \geq 1$. By Theorem 0.2, applied to the n columns of $\mathbf{X}_c{}^\top$,

$$\begin{aligned}
\text{rank}(\mathbf{X}_c) &= \text{rank}(\mathbf{X}_c{}^\top) && \text{(Theorem 0.4)} \\
&= n - \text{nullity}(\mathbf{X}_c{}^\top) && \text{(Theorem 0.2)} \\
&\leq n - 1. && (\text{nullity}(\mathbf{X}_c{}^\top) \geq 1)
\end{aligned}$$

□

Example 0.9 (Two points in three dimensions)**.** $\mathbf{X} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$ has $\bar{\underset{\sim}{x}} = (2, 2, 2)$, so

$$\mathbf{X}_c = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

The second row is -1 times the first, so $\text{rank}(\mathbf{X}_c) = 1 = n - 1$, and the column sums are zero: $\underset{\sim}{1}^\top \mathbf{X}_c = \underset{\sim}{0_3}^\top$.

Corollary 0.3 (A sample covariance matrix from $n \leq p$ points is singular)**.** *With $\mathbf{X}_c$ as in Theorem 0.6, the sample covariance matrix*

$$\mathbf{S} = \frac{1}{n-1}\,\mathbf{X}_c{}^\top \mathbf{X}_c$$

is $p \times p$ with $\text{rank}(\mathbf{S}) \leq n - 1$. So if $n \leq p$, then $\mathbf{S}$ is not invertible.

Proof

Proof.

$$\begin{aligned} \operatorname{rank}(\mathbf{S}) &= \operatorname{rank}(\mathbf{X}_c{}^{\top}\mathbf{X}_c) && \text{(multiplying by } \tfrac{1}{n-1} \neq 0 \text{ does not change the rank)} \\ &= \operatorname{rank}(\mathbf{X}_c) && \text{(Theorem 0.4)} \\ &\leq n-1. && \text{(Theorem 0.6)} \end{aligned}$$

If $n \leq p$, then $\operatorname{rank}(\mathbf{S}) \leq n-1 < p$, so $\mathbf{S}$ does not have full column rank, and it is not invertible by Theorem 1 in Diagonalization and Cholesky. □

Example 0.10 (As many points as dimensions)**.** The $n = 3$ points $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ in $p = 3$ dimensions have $\bar{\underset{\sim}{x}} = \frac{1}{3}(1,1,1)$, so

$$\mathbf{X}_c = \mathbf{I}_3 - \tfrac{1}{3}\,\underset{\sim}{1}\underset{\sim}{1}^{\top}, \qquad \mathbf{S} = \tfrac{1}{2}\,\mathbf{X}_c = \tfrac{1}{6}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix},$$

using $\mathbf{X}_c{}^{\top}\mathbf{X}_c = \mathbf{X}_c$ (the centering matrix is a symmetric projection). Then $\mathbf{S}\,\underset{\sim}{1} = \underset{\sim}{0}_3$ and $\operatorname{rank}(\mathbf{S}) = 2 = n - 1 < 3$. With one more point than dimensions ($n = p + 1$), the rank bound allows $\mathbf{S}$ to have full rank p.

Exercise 0.9 (Finding how large the rank of centered data can be)**.** Centering a data matrix subtracts the mean vector from every row, and Theorem 0.6 shows how large the rank of the result can be. Suppose that $\mathbf{X}$ holds 3 observations of 4 features, as a 3×4 matrix, and $\mathbf{X}_c$ is its centered version.
Find the number that, by Theorem 0.6, $\operatorname{rank}(\mathbf{X}_c)$ cannot exceed.

Hint

Compare the number from Theorem 0.6 with the one from Corollary 0.1.

Solution

Solution 0.9. By Theorem 0.6 with $n = 3$, $\operatorname{rank}(\mathbf{X}_c) \leq 3 - 1 = 2$. That number is smaller than $\min\{3, 4\} = 3$ from Corollary 0.1: the three centered rows add up to $\underset{\sim}{0}_4$, so they are linearly dependent.

Exercise 0.10 (Deciding whether a sample covariance matrix is invertible)**.** A study records 6 features for each of 4 people, and forms the 6×6 sample covariance matrix $\mathbf{S}$ defined in Corollary 0.3.

Decide whether $\mathbf{S}$ is invertible.

Hint

Compare n with p.

Solution

Solution 0.10. Here $n = 4$ and $p = 6$, so $n \leq p$. By Corollary 0.3, $\text{rank}(\mathbf{S}) \leq 4-1 = 3 < 6$, so $\mathbf{S}$ is not invertible.

Definition 0.2 (Affine subspace)**.** An **affine subspace** [6] of $\mathbb{R}^p$ is a set of the form

$$\{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{S}\}$$

for some vector $\underset{\sim}{x}_0 \in \mathbb{R}^p$ and some subspace $\mathcal{S}$ of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank): the image of $\mathcal{S}$ under the translation by $\underset{\sim}{x}_0$ (Definition 23 in Matrices).

Example 0.11 (A line that misses the origin is an affine subspace)**.** The line $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$ of Example 4 in Subspaces and Rank is not a subspace, but it is an affine subspace, with $\underset{\sim}{x}_0 = (0, 1)$ and $\mathcal{S} = \text{span}\{(1, 0)\}$ (Definition 5 in Subspaces and Rank):

$$\begin{aligned} (0,1) + c\,(1,0) &= (0 + c,\ 1 + 0) && \text{(Definition 9 in Matrices, Definition 7 in Vectors)} \\ &= (c, 1), && \text{(add)} \end{aligned}$$

and as c ranges over $\mathbb{R}$, $(c, 1)$ ranges over all of $\mathcal{T}$. Every subspace $\mathcal{S}$ is also an affine subspace, with $\underset{\sim}{x}_0 = \underset{\sim}{0}$.

Exercise 0.11 (Writing a line as an affine subspace)**.** A line such as $y = 2x + 3$ in $\mathbb{R}^2$ misses the origin, so it is not a subspace, but it is an affine subspace. Let $\mathcal{T} = \{(x, 2x + 3) : x \in \mathbb{R}\}$.
Find a vector $\underset{\sim}{x}_0$ and a subspace $\mathcal{S}$ with $\mathcal{T} = \{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{S}\}$.

Hint

Take $\underset{\sim}{x}_0$ to be the point of $\mathcal{T}$ with $x = 0$, and see what is left of $(x, 2x + 3)$ once you subtract it.

Solution

Solution 0.11. Take $\underset{\sim}{x}_0 = (0, 3)$, and let $\mathcal{S} = \text{span}\{(1, 2)\}$ be the span of $(1, 2)$. For any $c \in \mathbb{R}$,

$$(0, 3) + c\,(1, 2) = (c, 2c + 3),$$

so every $c \in \mathbb{R}$ gives a point $(c, 2c + 3)$ of $\mathcal{T}$, and every point of $\mathcal{T}$ arises this way.

Theorem 0.7 (A hyperplane is a translated subspace of dimension $p - 1$)**.** *Let $\underset{\sim}{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w}^\top \underset{\sim}{x} + b = 0\}$ be the hyperplane with normal vector $\underset{\sim}{w}$ and offset b (Definition 27 in Matrices), and let $\mathcal{H}_0 = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w}^\top \underset{\sim}{x} = 0\}$ be the hyperplane with the same normal vector and offset 0.*

1. *$\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank) of dimension $p - 1$ (Definition 9 in Subspaces and Rank).*
2. *$\mathcal{H}$ is not empty, and for any $\underset{\sim}{x}_0 \in \mathcal{H}$, $\mathcal{H} = \{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{H}_0\}$, so $\mathcal{H}$ is an affine subspace (Definition 0.2).*
3. *For any $\underset{\sim}{x}, \underset{\sim}{y} \in \mathcal{H}$, the normal vector is orthogonal to their difference: $\underset{\sim}{w} \perp (\underset{\sim}{x} - \underset{\sim}{y})$ (Definition 19 in Vectors).*

Proof

Proof. Here $\underset{\sim}{w}^\top$ is a $1 \times p$ matrix, and $\underset{\sim}{w}^\top \underset{\sim}{x} = \underset{\sim}{w} \cdot \underset{\sim}{x}$ (Example 8 in Matrices).
Part 1. $\mathcal{H}_0$ is the null space $\mathcal{N}(\underset{\sim}{w}^\top)$ (Definition 0.3), so it is a subspace (Theorem 0.2). The p columns of $\underset{\sim}{w}^\top$ are the numbers $w_1, \ldots, w_p$. Some $w_j \neq 0$, because $\underset{\sim}{w} \neq 0$, and that column on its own is linearly independent ($c\,w_j = 0$ forces $c = 0$), so $\text{rank}(\underset{\sim}{w}^\top) \geq 1$ (Definition 2 in Subspaces and Rank); and $\text{rank}(\underset{\sim}{w}^\top) \leq \min\{1, p\} = 1$ (Corollary 0.1). So

$$\begin{aligned} \dim(\mathcal{H}_0) &= \text{nullity}(\underset{\sim}{w}^\top) && \text{(Definition 0.1)} \\ &= p - \text{rank}(\underset{\sim}{w}^\top) && \text{(Theorem 0.2, for the } p \text{ columns of } \underset{\sim}{w}^\top) \\ &= p - 1. && \text{(the rank is 1)} \end{aligned}$$

Part 2. $\underset{\sim}{w} \cdot \underset{\sim}{w} = w_1^2 + \cdots + w_p^2 > 0$, because some $w_j \neq 0$. So $\underset{\sim}{x}^* \overset{\text{def}}{=} -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\,\underset{\sim}{w}$ is defined, and

$$\begin{aligned} \underset{\sim}{w}^\top \underset{\sim}{x}^* + b &= -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\,(\underset{\sim}{w}^\top \underset{\sim}{w}) + b && \text{(homogeneity of } \underset{\sim}{x} \mapsto \underset{\sim}{w}^\top \underset{\sim}{x}\text{, Theorem 12 in Matrices)} \\ &= -b + b && (\underset{\sim}{w}^\top \underset{\sim}{w} = \underset{\sim}{w} \cdot \underset{\sim}{w}) \\ &= 0, && \text{(arithmetic)} \end{aligned}$$

so $\underset{\sim}{x}^* \in \mathcal{H}$, and $\mathcal{H}$ is not empty. Now take any $\underset{\sim}{x}_0 \in \mathcal{H}$, so $\underset{\sim}{w}^\top \underset{\sim}{x}_0 = -b$. For any $\underset{\sim}{x} \in \mathbb{R}^p$,

$$\begin{aligned} \underset{\sim}{w}^\top(\underset{\sim}{x}-\underset{\sim}{x}_0) &= \underset{\sim}{w}^\top\underset{\sim}{x}-\underset{\sim}{w}^\top\underset{\sim}{x}_0 && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \underset{\sim}{w}^\top\underset{\sim}{x}-(-b) && (\underset{\sim}{x}_0\in\mathcal{H}) \\ &= \underset{\sim}{w}^\top\underset{\sim}{x}+b, && \text{(arithmetic)} \end{aligned}$$

so $\underset{\sim}{x}\in\mathcal{H}$ exactly when $\underset{\sim}{x}-\underset{\sim}{x}_0\in\mathcal{H}_0$, that is, exactly when $\underset{\sim}{x}=\underset{\sim}{x}_0+\underset{\sim}{s}$ with $\underset{\sim}{s}=\underset{\sim}{x}-\underset{\sim}{x}_0\in\mathcal{H}_0$. $\mathcal{H}_0$ is a subspace (part 1), so $\mathcal{H}$ has the form in Definition 0.2.

Part 3. For $\underset{\sim}{x},\underset{\sim}{y}\in\mathcal{H}$, both $\underset{\sim}{w}^\top\underset{\sim}{x}$ and $\underset{\sim}{w}^\top\underset{\sim}{y}$ equal $-b$, so

$$\begin{aligned} \underset{\sim}{w}\cdot(\underset{\sim}{x}-\underset{\sim}{y}) &= \underset{\sim}{w}^\top\underset{\sim}{x}-\underset{\sim}{w}^\top\underset{\sim}{y} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= (-b)-(-b) && (\underset{\sim}{x},\underset{\sim}{y}\in\mathcal{H}) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

□

Example 0.12 (The plane $x_1+x_2+x_3=1$ in $\mathbb{R}^3$)**.** Take $\underset{\sim}{w}=(1,1,1)$ and $b=-1$, so $\mathcal{H}=\{\underset{\sim}{x}:x_1+x_2+x_3-1=0\}$, the plane that Example 5 in Subspaces and Rank showed is not a subspace. By part 1, $\mathcal{H}_0=\{\underset{\sim}{x}:x_1+x_2+x_3=0\}$ is a subspace of dimension $3-1=2$.

- $\underset{\sim}{x}_0=(1,0,0)$ is in $\mathcal{H}$, since $1+0+0-1=0$, and so is $\underset{\sim}{y}=(0,1,0)$.
- Their difference $\underset{\sim}{y}-\underset{\sim}{x}_0=(-1,1,0)$ has entries summing to $-1+1+0=0$, so it is in $\mathcal{H}_0$, and $\underset{\sim}{y}=\underset{\sim}{x}_0+(-1,1,0)$, as part 2 says.
- $$\begin{aligned} \underset{\sim}{w}\cdot(-1,1,0) &= -1+1+0 \\ &= 0, \end{aligned}$$

  as part 3 says.
- The point built in the proof is

  $$\begin{aligned} \underset{\sim}{x}^* &= -\frac{-1}{3}(1,1,1) \\ &= (\tfrac{1}{3},\tfrac{1}{3},\tfrac{1}{3}), \end{aligned}$$

  and $\frac{1}{3}+\frac{1}{3}+\frac{1}{3}-1=0$.

Exercise 0.12 (Checking that a normal vector is orthogonal to a hyperplane)**.** Part 3 of Theorem 0.7 says that, for a hyperplane, its normal vector is orthogonal to the difference of any two of its points. Let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^3 : 2x_1 - x_2 + x_3 - 4 = 0\}$, with normal vector $\underset{\sim}{w} = (2, -1, 1)$. The points $\underset{\sim}{x} = (2, 0, 0)$ and $\underset{\sim}{y} = (0, 0, 4)$ are both in $\mathcal{H}$.
Check that $\underset{\sim}{w} \cdot (\underset{\sim}{x} - \underset{\sim}{y}) = 0$.

Hint

Compute $\underset{\sim}{x} - \underset{\sim}{y}$ first.

Solution

Solution 0.12. Here $\underset{\sim}{x} - \underset{\sim}{y} = (2, 0, -4)$, and

$$\underset{\sim}{w} \cdot (\underset{\sim}{x} - \underset{\sim}{y}) = 2 \cdot 2 + (-1) \cdot 0 + 1 \cdot (-4) = 0,$$

as part 3 of Theorem 0.7 says.

Theorem 0.8 (Where a hyperplane passes, and its shape in $\mathbb{R}^2$ and $\mathbb{R}^3$)**.** *Let* $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0\}$ *be a hyperplane (Definition 27 in Matrices).*

1. $\mathcal{H}$ *contains the origin* $\underset{\sim}{0}$ *exactly when* $b = 0$.
2. *If* $p = 2$*, then* $\mathcal{H}$ *is a line:* $\mathcal{H} = \{\underset{\sim}{x}_0 + t\,\underset{\sim}{v} : t \in \mathbb{R}\}$ *for some* $\underset{\sim}{x}_0 \in \mathbb{R}^2$ *and some nonzero* $\underset{\sim}{v} \in \mathbb{R}^2$.
3. *If* $p = 3$*, then* $\mathcal{H}$ *is a plane:* $\mathcal{H} = \{\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2 : s, t \in \mathbb{R}\}$ *for some* $\underset{\sim}{x}_0 \in \mathbb{R}^3$ *and some linearly independent* $\underset{\sim}{v}_1, \underset{\sim}{v}_2 \in \mathbb{R}^3$.

Proof

Proof. **Part 1.** Since $\underset{\sim}{w} \cdot \underset{\sim}{0} = 0$,

$$\begin{aligned} \underset{\sim}{w} \cdot \underset{\sim}{0} + b &= 0 + b && \text{(the dot product with } \underset{\sim}{0} \text{ is 0)} \\ &= b, && \text{(Theorem 4 in Convexity, Infimum and Sums)} \end{aligned}$$

so $\underset{\sim}{0} \in \mathcal{H}$ exactly when $b = 0$.
Parts 2 and 3. By Theorem 0.7, $\mathcal{H} = \{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{H}_0\}$ for any $\underset{\sim}{x}_0 \in \mathcal{H}$, where $\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ of dimension $p - 1$.
For $p = 2$, $\mathcal{H}_0$ has dimension 1, so it has a basis (Definition 6 in Subspaces and Rank) with one vector $\underset{\sim}{v}$. That vector is nonzero, and $\mathcal{H}_0 = \{t\,\underset{\sim}{v} : t \in \mathbb{R}\}$.
For $p = 3$, $\mathcal{H}_0$ has dimension 2, so it has a basis with two vectors $\underset{\sim}{v}_1$ and $\underset{\sim}{v}_2$. They are linearly independent, and $\mathcal{H}_0 = \{s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2 : s, t \in \mathbb{R}\}$. □

Example 0.13 (A line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$, checked on a grid of points)**.** In $\mathbb{R}^2$, take $\underset{\sim}{w} = (1, 1)$ and $b = -1$, the line of Example 24 in Matrices. The point $\underset{\sim}{x}_0 = -\frac{b}{\underset{\sim}{w}\cdot\underset{\sim}{w}}\,\underset{\sim}{w}$ is on it, and the vector $\underset{\sim}{v} = (-w_2, w_1)$ is nonzero with $\underset{\sim}{w}\cdot\underset{\sim}{v} = 0$, so it spans the subspace $\mathcal{H}_0$ of Theorem 0.7. In $\mathbb{R}^3$, take $\underset{\sim}{w} = (1, 1, 1)$ and $b = -1$, the plane of Example 0.12, with the independent vectors $\underset{\sim}{v}_1 = (1, -1, 0)$ and $\underset{\sim}{v}_2 = (0, 1, -1)$ in $\mathcal{H}_0$. The code evaluates $\underset{\sim}{w}\cdot\underset{\sim}{x} + b$ at the origin and at points $\underset{\sim}{x}_0 + t\,\underset{\sim}{v}$ and $\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2$.

```
level <- function(w, b, x) sum(w * x) + b

w2 <- c(1, 1)
b2 <- -1
point_line <- -b2 / sum(w2 * w2) * w2
v <- c(-w2[2], w2[1])
line_levels <- sapply(-2:2, function(t) level(w2, b2, point_line + t * v))

w3 <- c(1, 1, 1)
b3 <- -1
point_plane <- -b3 / sum(w3 * w3) * w3
v1 <- c(1, -1, 0)
v2 <- c(0, 1, -1)
grid <- expand.grid(s = -2:2, t = -2:2)
plane_levels <- mapply(
  function(s, t) level(w3, b3, point_plane + s * v1 + t * v2),
  grid$s, grid$t
)

c(
  origin_in_R2 = level(w2, b2, c(0, 0)),
  origin_in_R3 = level(w3, b3, c(0, 0, 0)),
  origin_with_b0 = level(w3, 0, c(0, 0, 0))
)
#>   origin_in_R2   origin_in_R3 origin_with_b0
#>             -1             -1              0
```

At the origin the left side equals the offset, so the origin is on the line only if b is 0: it gives -1 for the line and -1 for the plane, and 0 once the offset is changed to 0. All 5 points $\underset{\sim}{x}_0 + t\,\underset{\sim}{v}$ on the line and all 25 points $\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2$ on the plane satisfy the equation up to rounding error: 30 of the 30 points have $|\underset{\sim}{w}\cdot\underset{\sim}{x} + b| < 10^{-10}$.

Exercise 0.13 (Describing a line in $\mathbb{R}^2$ by a point and a vector)**.** Part 2 of Theorem 0.8 says that a hyperplane in $\mathbb{R}^2$ is a line $\{\underset{\sim}{x}_0 + t\,\underset{\sim}{v} : t \in \mathbb{R}\}$ for a point $\underset{\sim}{x}_0$ and a nonzero vector $\underset{\sim}{v}$. Let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^2 : 2x_1 + x_2 - 4 = 0\}$.
Find a point $\underset{\sim}{x}_0$ and a nonzero vector $\underset{\sim}{v}$ with $\mathcal{H} = \{\underset{\sim}{x}_0 + t\,\underset{\sim}{v} : t \in \mathbb{R}\}$.

> **Hint**
>
> Find one point of $\mathcal{H}$ by setting $x_2 = 0$, and take $\underset{\sim}{v} = (-w_2, w_1)$ as in Example 0.13.

Solution

Solution 0.13. Setting $x_2 = 0$ gives $x_1 = 2$, so $\underset{\sim}{x}_0 = (2, 0)$ is in $\mathcal{H}$. With $\underset{\sim}{w} = (2, 1)$, take $\underset{\sim}{v} = (-1, 2)$. Then $\underset{\sim}{x}_0 + t\,\underset{\sim}{v} = (2 - t, 2t)$, and

$$2\,(2 - t) + 2t - 4 = 0$$

for every t. Conversely, a point (x_1, x_2) of $\mathcal{H}$ has $x_2 = 4 - 2x_1$, so it is $\underset{\sim}{x}_0 + t\,\underset{\sim}{v}$ with $t = 2 - x_1$.

Theorem 0.9 (The graph of an affine function is a hyperplane)**.** *Let $\underset{\sim}{w} \in \mathbb{R}^p$, $b \in \mathbb{R}$. The graph of the function $\underset{\sim}{x} \mapsto \underset{\sim}{w} \cdot \underset{\sim}{x} + b$ is the set*

$$\left\{ \begin{bmatrix} \underset{\sim}{x} \\ y \end{bmatrix} \in \mathbb{R}^{p+1} : y = \underset{\sim}{w} \cdot \underset{\sim}{x} + b \right\}.$$

This set is a hyperplane in $\mathbb{R}^{p+1}$, with normal vector $\begin{bmatrix} \underset{\sim}{w} \\ -1 \end{bmatrix}$.

Proof

Proof. For $\begin{bmatrix} \underset{\sim}{x} \\ y \end{bmatrix} \in \mathbb{R}^{p+1}$,

$$\begin{aligned} y = \underset{\sim}{w} \cdot \underset{\sim}{x} + b &\iff \underset{\sim}{w} \cdot \underset{\sim}{x} - y + b = 0 && \text{(subtract } y \text{ from both sides)} \\ &\iff \begin{bmatrix} \underset{\sim}{w} \\ -1 \end{bmatrix} \cdot \begin{bmatrix} \underset{\sim}{x} \\ y \end{bmatrix} + b = 0. && \text{(dot product of stacked vectors)} \end{aligned}$$

So the set is the solution set of the equation of a hyperplane, with normal vector $\begin{bmatrix} \underset{\sim}{w} \\ -1 \end{bmatrix}$. That vector is nonzero, because its last entry is -1. □

Example 0.14 (The graph of a line is a line in the plane)**.** Take $p = 1$, $w = 2$, $b = 1$. The graph of $y = 2x + 1$ is the set of points (x, y) in $\mathbb{R}^2$ with $2x - y + 1 = 0$. By Theorem 0.9, this is a hyperplane with normal vector $(2, -1)$. The point $(0, 1)$ is on it, since $2 \cdot 0 - 1 + 1 = 0$. The point $(0, 0)$ is off it, since $2 \cdot 0 - 0 + 1 = 1 \neq 0$. Figure 1 shows the line, the two points, and the normal vector drawn from $(0, 1)$.

```
arrow_length <- grid::unit(0.15, "inches")
normal_arrow <- grid::arrow(length = arrow_length)
point_x <- c(0, 0)
point_y <- c(1, 0)
point_label <- c("on the line", "off the line")
points <- tibble::tibble(x = point_x, y = point_y, label = point_label)
normal <- tibble::tibble(x = 0, y = 1, xend = 2, yend = 0)
normal_aes <- ggplot2::aes(x = x, y = y, xend = xend, yend = yend)
x_limits <- c(-2, 3)
y_limits <- c(-2, 4)
ggplot2::ggplot(points) +
  ggplot2::aes(x = x, y = y, colour = label) +
  ggplot2::geom_abline(intercept = 1, slope = 2) +
  ggplot2::geom_segment(
    mapping = normal_aes,
    data = normal,
    arrow = normal_arrow,
    inherit.aes = FALSE
  ) +
  ggplot2::geom_point(size = 3) +
  ggplot2::coord_fixed(xlim = x_limits, ylim = y_limits) +
  ggplot2::labs(x = "x", y = "y", colour = NULL)
```

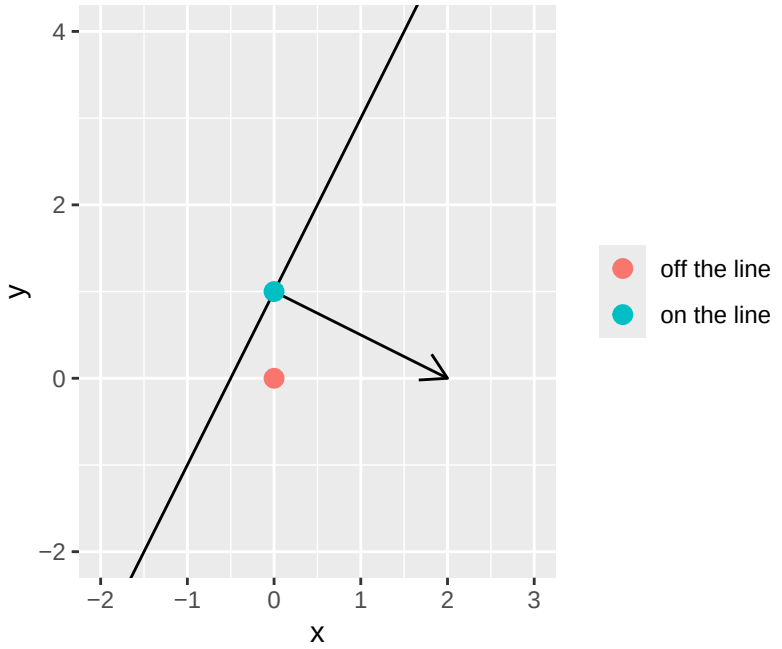


Figure 1: The graph of $y = 2x + 1$, a line in the plane, with its normal vector $(2, -1)$ drawn from the point $(0, 1)$ on it and the point $(0, 0)$ off it

Exercise 0.14 (Finding the normal vector of the graph of a regression surface)**.** Theorem 0.9 says that the graph of $\underset{\sim}{x} \mapsto \underset{\sim}{w} \cdot \underset{\sim}{x} + b$ is a hyperplane in $\mathbb{R}^{p+1}$. Let $f(x_1, x_2) = 3x_1 - x_2 + 2$, a regression surface with two features.
Find a normal vector for the graph of f in $\mathbb{R}^3$.

Hint

Read off $\underset{\sim}{w}$ and use the normal vector $\begin{bmatrix} \underset{\sim}{w} \\ -1 \end{bmatrix}$ from Theorem 0.9.

Solution

Solution 0.14. Here $\underset{\sim}{w} = (3, -1)$ and $b = 2$. By Theorem 0.9, the graph of f is a hyperplane in $\mathbb{R}^3$ with normal vector $(3, -1, -1)$: it holds the points (x_1, x_2, y) with $3x_1 - x_2 - y + 2 = 0$.

Definition 0.3 (Open half-space)**.** Let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0\}$ be a hyperplane (Definition 27 in Matrices). The two **open half-spaces** [7] of $\mathcal{H}$ are

$$\mathcal{H}^+ \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b > 0\} \quad \text{and} \quad \mathcal{H}^- \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b < 0\}. \tag{1}$$

Definition 0.4 (Closed half-space)**.** The **closed half-spaces** of the hyperplane $\mathcal{H}$ of Definition 0.3 also include $\mathcal{H}$ itself: $\mathcal{H}^+ \cup \mathcal{H}$ and $\mathcal{H}^- \cup \mathcal{H}$.

Remark

Remark. Replacing $(\underset{\sim}{w}, b)$ by $(-\underset{\sim}{w}, -b)$ describes the same hyperplane but swaps the names $\mathcal{H}^+$ and $\mathcal{H}^-$. The sign of $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ tells you which side of the hyperplane $\underset{\sim}{x}$ is on. A linear classifier uses this sign to assign one of two labels.

Example 0.15 (Half-spaces of a line in the plane)**.** Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\underset{\sim}{w} = (1, 2)$ and $b = -2$ (Definition 27 in Matrices), so $\mathcal{H} = \{\underset{\sim}{x} : x_1 + 2x_2 - 2 = 0\}$. Evaluate $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ at three points (Definition 0.3):

- At $(0, 0)$ it equals $0 + 0 - 2 = -2 < 0$, so $(0, 0) \in \mathcal{H}^-$.
- At $(2, 0)$ it equals $2 + 0 - 2 = 0$, so $(2, 0) \in \mathcal{H}$.
- At $(2, 1)$ it equals $2 + 2 - 2 = 2 > 0$, so $(2, 1) \in \mathcal{H}^+$.

Moving from $(2, 0)$ in the direction $\underset{\sim}{w} = (1, 2)$ gives $(3, 2)$, where the value is $3 + 4 - 2 = 5 > 0$. So the direction $\underset{\sim}{w}$ leads into $\mathcal{H}^+$.

Exercise 0.15 (Finding which open half-space holds a point)**.** The sign of $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ tells which open half-space of a hyperplane a point $\underset{\sim}{x}$ lies in. Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\underset{\sim}{w} = (2, -1)$ and $b = -1$.
Decide whether $(0, 0)$ is in $\mathcal{H}^+$ or in $\mathcal{H}^-$.

Hint

Evaluate $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ at $(0, 0)$.

Solution

Solution 0.15. At $(0, 0)$, $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0 + 0 - 1 = -1 < 0$, so $(0, 0)$ is in the open half-space $\mathcal{H}^-$.

Exercise 0.16 (Testing whether a point is in a closed half-space)**.** A closed half-space adds the hyperplane itself to an open half-space. Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\underset{\sim}{w} = (2, -1)$ and $b = -3$.
Decide whether the point $(2, 1)$ is in the closed half-space $\mathcal{H}^+ \cup \mathcal{H}$.

Hint

Evaluate $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ at $(2, 1)$.

Solution

Solution 0.16. At $(2,1)$, $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 4 - 1 - 3 = 0$, so $(2,1) \in \mathcal{H}$. It is therefore in the closed half-space $\mathcal{H}^+ \cup \mathcal{H}$, though not in the open half-space $\mathcal{H}^+$.

Remark (The two sides of a hyperplane)

Remark 0.1 (The two sides of a hyperplane). Every point of $\mathbb{R}^p$ is in exactly one of $\mathcal{H}^+$, $\mathcal{H}$ and $\mathcal{H}^-$, because the number $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ is positive, zero or negative. The vector $\underset{\sim}{w}$ points into $\mathcal{H}^+$: the number $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ goes up when $\underset{\sim}{x}$ moves in the direction $\underset{\sim}{w}$.

Theorem 0.10 (Differences of points in a hyperplane and in a half-space)**.** *Let $\underset{\sim}{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, and let $\mathcal{H}$, $\mathcal{H}^+$ be the hyperplane and open half-space of Definition 0.3. For a set $S \subseteq \mathbb{R}^p$, let $D(S) \stackrel{def}{=} \{\underset{\sim}{y} - \underset{\sim}{x} : \underset{\sim}{x}, \underset{\sim}{y} \in S\}$ be the set of differences of its points.*

1. *$D(\mathcal{H}) = \{\underset{\sim}{s} : \underset{\sim}{w} \cdot \underset{\sim}{s} = 0\}$, a subspace of dimension $p - 1$.*
2. *$D(\mathcal{H}^+) = \mathbb{R}^p$, a subspace of dimension p.*

The same holds for $\mathcal{H}^-$.

Proof

Proof. **Part 1.** Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathcal{H}$. By Theorem 0.7 part 3, $\underset{\sim}{w} \cdot (\underset{\sim}{y} - \underset{\sim}{x}) = 0$, so $\underset{\sim}{y} - \underset{\sim}{x} \in \mathcal{H}_0 = \{\underset{\sim}{s} : \underset{\sim}{w} \cdot \underset{\sim}{s} = 0\}$. Conversely, take any $\underset{\sim}{x}_0 \in \mathcal{H}$ (Theorem 0.7 part 2 gives such a point). For any $\underset{\sim}{s} \in \mathcal{H}_0$, the point $\underset{\sim}{x}_0 + \underset{\sim}{s}$ is in $\mathcal{H}$ by the same part, and $(\underset{\sim}{x}_0 + \underset{\sim}{s}) - \underset{\sim}{x}_0 = \underset{\sim}{s}$. So $D(\mathcal{H}) = \mathcal{H}_0$, which has dimension $p - 1$ by Theorem 0.7 part 1.
Part 2. Every difference is in $\mathbb{R}^p$, so $D(\mathcal{H}^+) \subseteq \mathbb{R}^p$. For the reverse inclusion, take any $\underset{\sim}{x}_0 \in \mathcal{H}$. Take any $\underset{\sim}{v} \in \mathbb{R}^p$. For a number $t > 0$ to be chosen, let $\underset{\sim}{x}_t \stackrel{\text{def}}{=} \underset{\sim}{x}_0 + t\,\underset{\sim}{w}$. Then

$$\begin{aligned} \underset{\sim}{w} \cdot \underset{\sim}{x}_t + b &= (\underset{\sim}{w} \cdot \underset{\sim}{x}_0 + b) + t\,\underset{\sim}{w} \cdot \underset{\sim}{w} && \text{(linearity of } \underset{\sim}{x} \mapsto \underset{\sim}{w} \cdot \underset{\sim}{x}) \\ &= t\,\underset{\sim}{w} \cdot \underset{\sim}{w}, && (\underset{\sim}{x}_0 \in \mathcal{H}) \end{aligned}$$

which is positive, so $\underset{\sim}{x}_t \in \mathcal{H}^+$. Next,

$$\begin{aligned} \underset{\sim}{w} \cdot (\underset{\sim}{x}_t + \underset{\sim}{v}) + b &= (\underset{\sim}{w} \cdot \underset{\sim}{x}_t + b) + \underset{\sim}{w} \cdot \underset{\sim}{v} && \text{(linearity of } \underset{\sim}{x} \mapsto \underset{\sim}{w} \cdot \underset{\sim}{x}) \\ &= t\,\underset{\sim}{w} \cdot \underset{\sim}{w} + \underset{\sim}{w} \cdot \underset{\sim}{v}. && \text{(the display above)} \end{aligned}$$

Because $\underset{\sim}{w} \cdot \underset{\sim}{w} > 0$, this is positive once $t > -\underset{\sim}{w} \cdot \underset{\sim}{v} / \underset{\sim}{w} \cdot \underset{\sim}{w}$. Choose such a $t > 0$. Then $\underset{\sim}{x}_t$

and $\underset{\sim}{x}_t + \underset{\sim}{v}$ are both in $\mathcal{H}^+$, and their difference is $\underset{\sim}{v}$. So $\underset{\sim}{v} \in D(\mathcal{H}^+)$. The whole space $\mathbb{R}^p$ is a subspace of dimension p (Definition 9 in Subspaces and Rank).
The proof for $\mathcal{H}^-$ is the same with $-\underset{\sim}{w}$ and $-b$ in place of $\underset{\sim}{w}$ and b. □

Example 0.16 (Two iris species on either side of a line)**.** The `iris` data set in R has the petal length and petal width, in centimeters, of 150 irises. Take the 100 flowers of the species versicolor and virginica, and let $\underset{\sim}{x} = (x_1, x_2)$ be the petal length and petal width of one flower. A logistic regression of the species on $\underset{\sim}{x}$ gives a coefficient vector $\underset{\sim}{w}$ and an intercept b. The line $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0$ is a hyperplane in $\mathbb{R}^2$ (Definition 27 in Matrices), and the two species mostly fall in its two open half-spaces (Definition 0.3). Figure 2 shows the line and the flowers.

```
two_species <- dplyr::filter(iris, Species != "setosa") |> droplevels()
fit <- glm(
  Species ~ Petal.Length + Petal.Width,
  family = binomial,
  data = two_species
)
w <- coef(fit)[c("Petal.Length", "Petal.Width")]
b <- coef(fit)[["(Intercept)"]]
two_species$score <- as.vector(
  as.matrix(two_species[c("Petal.Length", "Petal.Width")]) %*% w + b
)
two_species$side <- ifelse(two_species$score > 0, "positive", "negative")
ggplot2::ggplot(two_species) +
  ggplot2::aes(x = Petal.Length, y = Petal.Width, colour = Species) +
  ggplot2::geom_point() +
  ggplot2::geom_abline(intercept = -b / w[[2]], slope = -w[[1]] / w[[2]]) +
  ggplot2::labs(x = "Petal length (cm)", y = "Petal width (cm)")
```

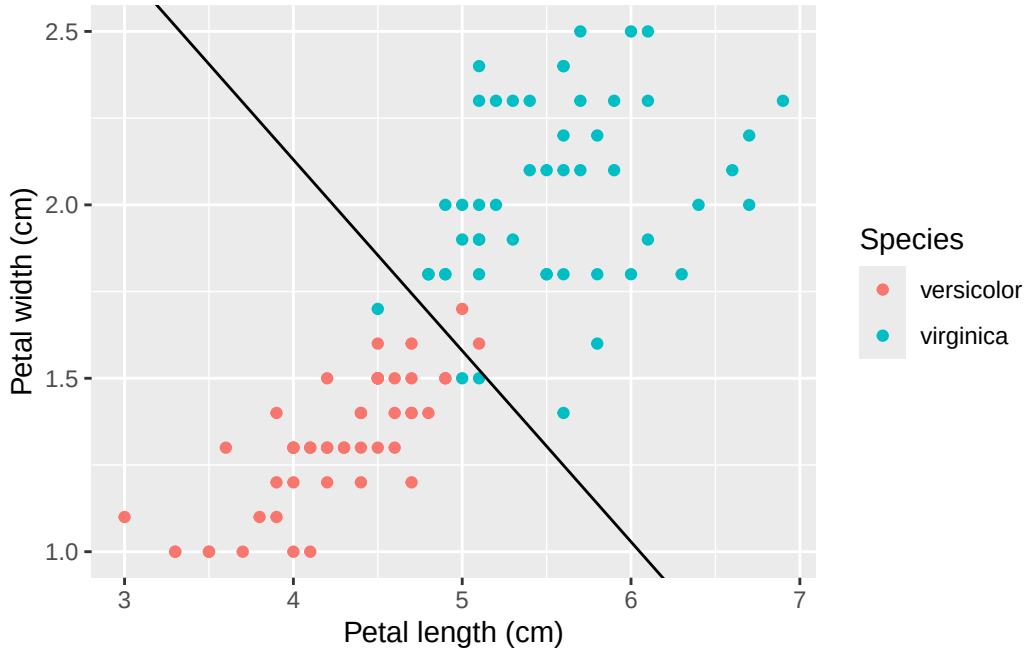


Figure 2: Petal length and width of versicolor and virginica irises, with the line where a logistic regression is indifferent between the two species

```
table(two_species$Species, two_species$side)
#>
#>              negative positive
#>   versicolor       47        3
#>   virginica         3       47
```

The vector $\underset{\sim}{w}$ points toward the virginica side, so the positive half-space $\mathcal{H}^+$ holds 50 of the 100 flowers, and 47 of them are virginica. The line is a set of dimension $p-1=1$, and each side is a region of the plane of dimension $p=2$ (Theorem 0.10).

Exercise 0.17 (Which side of the plane?)**.** Let $\mathcal{H}$ be the plane in $\mathbb{R}^3$ with $\underset{\sim}{w}=(1,1,1)$ and $b=-1$ (Example 0.12).

1. For each point, say whether it is in $\mathcal{H}^+$, $\mathcal{H}$ or $\mathcal{H}^-$ (Definition 0.3): $(0,0,0)$, $(1,0,0)$ and $(1,1,1)$.
2. Write the plane and its half-spaces using $-\underset{\sim}{w}$ and $-b$. Which points from part 1 change sides?
3. State the dimension of $D(\mathcal{H})$ and of $D(\mathcal{H}^+)$ (Theorem 0.10).

Solution

Solution 0.17.

1. Compute $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = x_1 + x_2 + x_3 - 1$ at each point.
   - At $(0,0,0)$ it equals $0+0+0-1 = -1 < 0$, so the point is in $\mathcal{H}^-$.
   - At $(1,0,0)$ it equals $1+0+0-1 = 0$, so the point is in $\mathcal{H}$.
   - At $(1,1,1)$ it equals $1+1+1-1 = 2 > 0$, so the point is in $\mathcal{H}^+$.
2. With $-\underset{\sim}{w} = (-1,-1,-1)$ and $-b = 1$, the equation is $-x_1 - x_2 - x_3 + 1 = 0$. Multiplying both sides by -1 gives the original equation, so it is the same plane. The new expression is the negative of the old one at every point, so the new positive half-space is the old $\mathcal{H}^-$ and the new negative half-space is the old $\mathcal{H}^+$. The origin $(0,0,0)$ is now in the positive half-space, and $(1,1,1)$ is now in the negative half-space. The point $(1,0,0)$ stays on the plane. So the origin and $(1,1,1)$ both change names, and the point on the plane does not.
3. Here $p = 3$. $D(\mathcal{H})$ is the subspace $\{\underset{\sim}{s} : s_1 + s_2 + s_3 = 0\}$, which has dimension $p - 1 = 2$. $D(\mathcal{H}^+) = \mathbb{R}^3$ has dimension $p = 3$.

Definition 0.5 (Rank factorization)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with $\operatorname{rank}(\mathbf{A}) = r \geq 1$. A **rank factorization** [8] of $\mathbf{A}$ is a product

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times r}{\mathbf{C}}\ \underset{r\times n}{\mathbf{R}}\ .$$

Example 0.17 (A rank factorization of a rank-one matrix)**.** The matrix $\mathbf{A}$ in Example 0.1 has rank 1 (Example 0.1), and

$$\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}1 & -2 & -2\end{bmatrix} = \begin{bmatrix}1\cdot 1 & 1\cdot(-2) & 1\cdot(-2)\\ 3\cdot 1 & 3\cdot(-2) & 3\cdot(-2)\end{bmatrix} = \begin{bmatrix}1 & -2 & -2\\ 3 & -6 & -6\end{bmatrix},$$

so $\mathbf{C} = \begin{bmatrix}1\\3\end{bmatrix}$ (2×1) and $\mathbf{R} = \begin{bmatrix}1 & -2 & -2\end{bmatrix}$ (1×3) are a rank factorization. It is not unique: $\mathbf{C} = \begin{bmatrix}2\\6\end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix}\frac{1}{2} & -1 & -1\end{bmatrix}$ give the same product. A product $\mathbf{CR}$ with $\mathbf{C}$ of size 2×2 is not a rank factorization of this $\mathbf{A}$, because the inner dimension must equal $\operatorname{rank}(\mathbf{A}) = 1$.

Theorem 0.11 (Every nonzero matrix has a rank factorization)**.** *Every $m \times n$ matrix $\mathbf{A}$ with $\operatorname{rank}(\mathbf{A}) = r \geq 1$ has a rank factorization (Definition 0.5). One is given by taking the columns of $\mathbf{C}$ to be any basis of $\mathcal{C}(\mathbf{A})$.*

Proof

Proof. $\mathcal{C}(\mathbf{A})$ has dimension r (Theorem 0.1), so it has a basis $\underset{\sim}{c}_1, \ldots, \underset{\sim}{c}_r$ (Theorem 9 in Subspaces and Rank); let $\mathbf{C}$ be the $m \times r$ matrix with these columns. Each column $\underset{\sim}{a}_j$ of $\mathbf{A}$ is in $\mathcal{C}(\mathbf{A})$, so $\underset{\sim}{a}_j = r_{1j}\underset{\sim}{c}_1 + \cdots + r_{rj}\underset{\sim}{c}_r$ for some numbers r_{ij}, because the basis spans $\mathcal{C}(\mathbf{A})$. Let $\mathbf{R}$ be the $r \times n$ matrix with entries r_{ij}, whose column j is $\underset{\sim}{r}_j = (r_{1j}, \ldots, r_{rj})$. Then column j of $\mathbf{CR}$ is

$$\begin{aligned} \mathbf{C}\underset{\sim}{r}_j &= r_{1j}\underset{\sim}{c}_1 + \cdots + r_{rj}\underset{\sim}{c}_r && \text{(Theorem 3 in Subspaces and Rank)} \\ &= \underset{\sim}{a}_j, && \text{(choice of the } r_{ij}\text{)} \end{aligned}$$

so $\mathbf{CR} = \mathbf{A}$. □

Example 0.18 (Building a rank factorization from a basis)**.** For $\mathbf{A}$ in Example 0.1, $(1, 3)$ spans $\mathcal{C}(\mathbf{A})$ (Example 0.2), and it is linearly independent because it is nonzero ($c\,(1, 3) = \underset{\sim}{0}_2$ forces $c = 0$), so it is a basis of $\mathcal{C}(\mathbf{A})$. The columns of $\mathbf{A}$ are $(1, 3) = 1 \cdot (1, 3)$, $(-2, -6) = -2 \cdot (1, 3)$ and $(-2, -6) = -2 \cdot (1, 3)$, so the construction in Theorem 0.11 gives $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$, the factorization in Example 0.17.

Exercise 0.18 (Finding a rank factorization of a rank-one matrix)**.** A rank factorization writes an $m \times n$ matrix with rank r as a matrix product of an $m \times r$ and an $r \times n$ matrix. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}.$$

Find a rank factorization of $\mathbf{A}$.

Hint

In $\mathbf{A}$, the second column is twice the first, so $\operatorname{rank}(\mathbf{A}) = 1$.

Solution

Solution 0.18. Columns of $\mathbf{A}$ are $(2, 1)$ and $(4, 2) = 2\,(2, 1)$, so $\operatorname{rank}(\mathbf{A}) = 1$, and

$$\begin{bmatrix}2\\1\end{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix} = \begin{bmatrix}2\cdot 1 & 2\cdot 2\\ 1\cdot 1 & 1\cdot 2\end{bmatrix} = \begin{bmatrix}2 & 4\\ 1 & 2\end{bmatrix} = \mathbf{A}.$$

So $\mathbf{C} = \begin{bmatrix}2\\1\end{bmatrix}$ (2×1) and $\mathbf{R} = \begin{bmatrix}1 & 2\end{bmatrix}$ (1×2) are a rank factorization.

Exercise 0.19 (Building a rank factorization from a basis of the column space)**.** Theorem 0.11 builds a rank factorization from a basis of the column space. Let

$$\mathbf{A} = \begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 6\\ 1 & 0 & 1\end{bmatrix},$$

in which column 3 is column 1 plus column 2.
Find a rank factorization $\mathbf{A} = \mathbf{CR}$ in which $\mathbf{C}$ holds columns 1 and 2 of $\mathbf{A}$.

Hint

Column j of $\mathbf{R}$ says how much of column 1 and of column 2 of $\mathbf{C}$ add up to column j of $\mathbf{A}$.

Solution

Solution 0.19. Columns 1 and 2 of $\mathbf{A}$, $(1,2,1)$ and $(2,4,0)$, are linearly independent: the third entry of $c_1\,(1,2,1) + c_2\,(2,4,0)$ is c_1, and then the first is $2c_2$. They span $\mathcal{C}(\mathbf{A})$, since the third column is the first plus the second, so $\operatorname{rank}(\mathbf{A}) = 2$. Each column of $\mathbf{A}$ in terms of them:

$$\begin{aligned}(1,2,1) &= 1\cdot(1,2,1) + 0\cdot(2,4,0),\\ (2,4,0) &= 0\cdot(1,2,1) + 1\cdot(2,4,0),\\ (3,6,1) &= 1\cdot(1,2,1) + 1\cdot(2,4,0).\end{aligned}$$

So

$$\mathbf{C} = \begin{bmatrix}1 & 2\\ 2 & 4\\ 1 & 0\end{bmatrix}, \qquad \mathbf{R} = \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix},$$

and multiplying gives back $\mathbf{A}$.

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[8] Wikipedia contributors, "Rank factorization — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Rank_factorization